

Multiplicity Operator Calculus: Prime-Indexed Dynamics on Hypergraphs

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Abstract

Multiplicity Operator Calculus (MOC) formalizes a non-reductive, prime-indexed framework for modeling interdependent processes on hypergraphs. Inspired by the recursive epistemologies of Eastern mathematics, MOC treats number not as abstraction but as relational operator. Each prime p defines a family of noncommuting operators $\hat{P}_p$ that act simultaneously on nodes and relations, generating nested cycles of resonance and transformation. The calculus unites rhythmic, geometric, and symbolic domains by encoding process structure through factorization: composition $\prod \hat{P}_p^{k_p}$ represents both algebraic and temporal layering. Unlike linear differential or matrix approaches, MOC prioritizes recursive flows over static state evolution, enabling the modeling of systems whose coherence depends on rhythmic resonance rather than equilibrium. Applications span symbolic computation, musical architecture, prime-based synchronization networks, and ethical or cognitive dynamics in recursive agent systems. This paper develops the algebraic rules, commutation relations, and resonance functionals defining the MOC and demonstrates its use through a 108-cycle operator construct embodying binary–ternary interplay.

Keywords: multiplicity, hypergraph dynamics, prime operators, resonance, recursion, noncommutative systems

Executive Summary

Problem Statement and Motivation

Classical models privilege objects and linear evolution; relations are secondary and often fixed. Many real systems—symbolic, musical, cognitive, ethical—exhibit coherence through nested

cycles and context-sensitive coupling that standard differential or matrix methods capture only indirectly. We seek a calculus where *relations* and *rhythmic structure* are primary, and where number encodes process rather than quantity.

Core Contribution

We introduce the *Multiplicity Operator Calculus* (MOC): a prime-indexed, noncommutative operator framework acting on states and hyperedges of a hypergraph. For each prime p , a family $\{\hat{P}_p\}$ generates subdivision, rotation, accent, and permutation modes whose compositions realize nested periodicities. Validation is *resonance-based*: a functional

$$R(H, \{\hat{P}\}; D) \in [0, 1]$$

measures fit between modeled structure and target data D (e.g., cycles in time series, phrase boundaries, or symbolic constraints). As an exemplar, we construct a full $n = 108 = 2^2 \cdot 3^3$ operator word that integrates ternary phrasing with binary micro-pulse.

Key Results and Applications

- **Algebra:** Closed rules for composition, lifting across prime powers, and cross-prime braid relations; normal forms up to phase equivalence.
- **Resonance:** Practical criteria and algorithms to maximize R , enabling “proof by resonance” for pattern adequacy.
- **108-cycle:** A transparent build demonstrating how factorization sets macro–micro coupling and how operator order controls feel.
- **Applications:** (i) symbolic computation via rewrite orchestration on hypergraphs; (ii) musical architecture and metrical design; (iii) synchronization networks with prime-gated stability; (iv) cognitive/ethical policy constraints modeled as conserved quantities under $\hat{P}_p$.

Roadmap of the Document

Section 1 formalizes state spaces, hypergraph structure, and operator families. Section 2 develops commutation, lifting, and braid relations with illustrative lemmas. Section 3 defines the resonance functional and optimization procedures. Section 4 presents the 108-cycle construction, ablation studies on operator order, and empirical resonance profiles. Section 5 surveys applications and gives implementation notes. Section 6 concludes with open problems and directions for comparative studies with classical spectral and automata-theoretic methods.

Philosophical and Historical Context

Eastern Mathematics: Recursion, Relational Ontology, and Sutra Logic

In the intellectual traditions of India, China, and East Asia, mathematics evolved within a relational and recursive ontology rather than a mechanistic one. Number was conceived not as abstraction but as expression of interdependence. The Vedic and Buddhist doctrines of *Pratītyasamutpāda* (dependent origination) and the Taoist notion of the *Ten Thousand Things* unfolding from the *One* describe multiplicity as an emergent unity: relations produce entities, not the reverse.

Mathematical reasoning in these contexts operated through *sutra logic*—compact aphorisms encoding recursive procedures. Each sutra functioned simultaneously as algorithm, mnemonic, and cosmological statement. Knowledge was revealed through iterative interpretation, not static proof. Recursion served as epistemic method and ontological model.

Multiplicity Versus Reduction: Relations-First Stance

Western formalism tends toward reduction: decomposing wholes into independent parts and reassembling them through linear rules. Multiplicity theory in the Eastern sense reverses this orientation. It begins with networks of mutual conditioning where nodes have no existence apart from their connective roles. This relations-first stance treats mathematics as a mirror of dynamic balance rather than a tool of dissection.

Within this view, *multiplicity* is not numerical excess but structured co-arising. A number such as $108 = 2^2 \cdot 3^3$ is not a mere product; it encodes nested rhythmic and symbolic layers. Prime factors act as archetypal generators of relational order.

Symbolic Cycles in Practice: Calendar, Breath, Music, and Ritual

Traditional applications of Eastern mathematical ideas are found in cyclical systems. Calendrical computation linked celestial periodicities with ritual timing. Yogic and Daoist practices mapped breathing rhythms and bodily cycles to integer ratios. Classical Indian and East Asian music employed binary and ternary meter interlocks, structurally analogous to prime decomposition. Temple geometry, mandala construction, and mantra recitation all used numerical proportion as a mode of synchronization between microcosm and macrocosm.

These practices embody recursion physically: repetition with modulation, symmetry with deviation, and invariance through transformation. Mathematical order thus operated as a lived rhythm, not an abstract invariant.

Operationalization in This Report

The *Multiplicity Operator Calculus* formalizes these principles in algebraic and computational language. Primes become operators $\hat{P}_p$ acting recursively on hypergraphs; factorization corresponds to hierarchical cycles. Resonance replaces proof: coherence is verified when generated patterns align with empirical or symbolic cycles.

This report translates the philosophical motifs of recursion, interdependence, and cyclic symmetry into a rigorous operator framework. By embedding the relational ontology of Eastern mathematics into prime-indexed dynamics, it unites metaphysical intuition with executable formalism—an explicit bridge between symbolic cosmology and algorithmic modeling.

Preliminaries and Notation

Time Lattice

Let the discrete cyclic time domain be

$$\mathbb{T}_n = \mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n-1\},$$

which serves as the base lattice for periodic processes of length n . Arithmetic on $\mathbb{T}_n$ is taken modulo n . Each tick $t \in \mathbb{T}_n$ represents a phase position within a closed cycle.

Signals and Patterns

A *signal* or *pattern* is a function

$$x : \mathbb{T}_n \rightarrow \mathbb{R}^k,$$

where k denotes the number of feature channels (e.g., amplitude, color, intensity, or symbolic value). The space of all such signals is $\mathcal{X}_n = \{x : \mathbb{T}_n \rightarrow \mathbb{R}^k\}$, equipped with pointwise addition and scalar multiplication, forming a real vector space.

Hypergraph Structure

The relational substrate of multiplicity is modeled as a hypergraph

$$H = (V, E, \iota),$$

where V is the set of vertices (states), E the set of hyperedges (relations), and $\iota : E \rightarrow \mathcal{P}(V)$ the incidence map assigning to each edge its participating vertices. Unlike a standard graph, hyperedges may connect any number of vertices, representing higher-order relations.

Each vertex $v \in V$ and hyperedge $e \in E$ carries state attachments

$$\sigma : V \rightarrow \mathbb{R}^k, \quad \tau : E \rightarrow \mathbb{R}^m,$$

encoding local and relational attributes. Operators will act jointly on (σ, τ) , allowing transformation of both nodes and the relations that bind them.

Operator Words and Composition Order

An *operator word* is an ordered product of prime-indexed operators:

$$\mathcal{W} = \hat{O}_m \hat{O}_{m-1} \cdots \hat{O}_1, \quad \hat{O}_i \in \{\hat{P}_p^{(r)} : p \text{ prime}, r \geq 1\}.$$

Application is right-to-left:

$$x' = \mathcal{W}[x] = \hat{O}_m(\hat{O}_{m-1}(\cdots \hat{O}_1(x))).$$

Noncommutativity implies that reordering the same multiset of primes generally produces distinct outcomes. The set of all operator words with the same prime content forms an equivalence class under resonance equivalence, denoted $[\mathcal{W}]_R$.

Divisibility Indicator and Class Labels

For integer divisibility, define the indicator function

$$[d \mid t] = \begin{cases} 1, & \text{if } t \equiv 0 \pmod{d}, \\ 0, & \text{otherwise.} \end{cases}$$

This binary function allows the construction of hierarchical accent and phase structures. Given a cycle length n and a set of divisors $\{d_i\} \subseteq \mathbb{N}$, we may assign each tick $t \in \mathbb{T}_n$ a *class label*

$$C(t) = \{d_i : [d_i \mid t] = 1\},$$

representing its membership in overlapping rhythmic or symbolic strata. For example, in $n = 108$, the classes corresponding to $d = 2, 3, 9, 27$ define binary and ternary accent layers.

These primitives provide the syntactic foundation for the Multiplicity Operator Calculus: signals live on the lattice $\mathbb{T}_n$, relations inhabit H , and operator words encode recursive transformations whose resonance properties are analyzed in later sections.

Multiplicity Space and Carriers

Definition of the Multiplicity Space

The fundamental carrier of the Multiplicity Operator Calculus is the *multiplicity space*

$$\mathcal{M} = (\mathcal{X}, H),$$

where $\mathcal{X}$ denotes the signal space $\mathcal{X}_n = \{x : \mathbb{T}_n \rightarrow \mathbb{R}^k\}$ and $H = (V, E, \iota)$ is the hypergraph of relational topology introduced previously.

Elements of $\mathcal{M}$ therefore encode both *local states* and *relational structure*. Operators $\hat{P}_p^{(r)}$ act simultaneously on these two layers:

$$\hat{P}_p^{(r)} : (\mathcal{X}, H) \longrightarrow (\mathcal{X}', H'),$$

where $\mathcal{X}'$ and H' denote the updated state and hypergraph under a prime-indexed transformation (e.g., subdivision, rotation, or permutation). Each $\mathcal{M}$ is thus a dynamic carrier—its internal organization may change while preserving coherence through resonance invariants.

Automorphisms and Gauge Perspective

The internal symmetries of the multiplicity space form the automorphism group

$$\text{Aut}(\mathcal{M}) = \{ (\phi_V, \phi_E) \mid \phi_V : V \rightarrow V, \phi_E : E \rightarrow E, \iota \circ \phi_E = \mathcal{P}(\phi_V) \circ \iota \},$$

where $\mathcal{P}(\phi_V)$ denotes the induced action on subsets of V . These automorphisms represent relabelings of vertices and edges that leave the incidence structure invariant.

From a gauge perspective, each automorphism corresponds to a local symmetry transformation of the multiplicity field. If (σ, τ) are the vertex and edge states, an automorphism acts as

$$(\sigma, \tau) \mapsto (g_V \circ \sigma, g_E \circ \tau),$$

with g_V, g_E drawn from a local gauge group G (often $G \subseteq \text{GL}(k, \mathbb{R})$). The MOC treats these symmetries as redundancies of description—different coordinate expressions of the same relational configuration.

Gauge transformations commuting with all $\hat{P}_p^{(r)}$ define the *stabilizer subgroup* $\text{Stab}_G(\mathcal{M})$, corresponding to conserved relational frames.

Invariants and Conservation Constraints

Within each multiplicity space, certain scalar quantities remain invariant under admissible operator action. These invariants encode conservation of structure and ethics of transformation:

- **Energy Invariant:**

$$E[x] = \sum_{t \in \mathbb{T}_n} \|x(t)\|^2.$$

Preserved under norm-preserving operators (rotations, permutations).

- **Area or Volume Invariant:** For geometric realizations, define

$$A[\sigma, \tau] = \sum_{e \in E} \text{Vol}(\iota(e)),$$

which measures conserved relational “extent.” Sulba-style operators ($\hat{P}_p$ corresponding to cut-and-join) maintain A constant.

- **Fairness or Ethical Invariant:** For systems representing allocations or information flows,

$$F[\sigma] = \min_i \sigma_i / \max_i \sigma_i,$$

quantifies balance among vertices. Operator sets satisfying $F[\sigma'] \geq F[\sigma]$ are deemed ethically admissible.

These invariants act as *gauge constraints* on the permissible dynamics within $\mathcal{M}$. Resonant evolution respects them automatically: only operator compositions that preserve or restore these measures yield high resonance $R(\mathcal{M})$. Thus, the multiplicity space is not merely a passive container for transformation—it is an active metric field enforcing harmony between algebraic action and relational integrity.

Prime-Indexed Operator Calculus (MOC)

Operator Families on States

Fix a cycle length n . All state operators act on $x \in \mathcal{X}_n = \{x : \mathbb{T}_n \rightarrow \mathbb{R}^k\}$. Admissibility: S_p , R_{p^r} , A_{p^r} are defined when $p^r \mid n$; otherwise treat them as virtual refinements with lift–project (omitted here for brevity).

Subdivision S_p . Lifts $\mathcal{X}_n \rightarrow \mathcal{X}_{pn}$:

$$(S_p x)(pt + u) = x(t), \quad t \in \mathbb{T}_n, u \in \{0, \dots, p-1\}.$$

Accent $A_{p^r}^\alpha$. Adds a level- p^r gate (or multiplies by a weight). Additive form:

$$(A_{p^r}^\alpha x)(t) = x(t) + \alpha [p^r \mid t] e_1,$$

with $[d \mid t] \in \{0, 1\}$ the divisibility indicator and e_1 a chosen channel.

Rotation $\mathbf{R}_{p^r}^\phi$. Cyclic shift aligned to the p^r -grid:

$$(\mathbf{R}_{p^r}^\phi x)(t) = x\left(t + \phi \cdot \frac{n}{p^r}\right), \quad \phi \in \mathbb{Z}.$$

Permutation $\mathbf{W}_p^\pi$. Reorders the p subticks inside each parent cell after $\mathbf{S}_p$. For $\pi \in S_p$,

$$(\mathbf{W}_p^\pi x)(pt + u) = x(pt + \pi(u)).$$

Relation Operators on H

Let $H = (V, E, \iota)$. Relation operators $\hat{Q}_p$ act on incidence while coordinating state attachments (σ, τ) .

Split $\hat{Q}_p^{\text{split}}$. Each $e \in E$ becomes p siblings $e^{(u)}$ with refined incidence:

$$\iota'(e^{(u)}) = \iota(e), \quad u = 0, \dots, p-1, \quad \tau'(e^{(u)}) = \text{Refine}_p(\tau(e), u).$$

Merge $\hat{Q}_p^{\text{merge}}$. Inverse of split on coherent siblings: p edges with identical incidence collapse to one, τ aggregated by a conserved rule.

Fold $\hat{Q}_p^{\text{fold}}$. Quotients a p -cyclic arrangement of vertices/edges by rotation, enforcing periodic boundary conditions.

Relabel $\hat{Q}_p^{\text{rel}}$. Applies an incidence-preserving permutation $(\phi_V, \phi_E) \in \text{Aut}(H)$ constrained to p -cycles.

Prime Word and Composite Word

For each prime p and level $r \geq 1$, define a *prime word*

$$\hat{P}^{(p)} = \mathbf{W}_p^{\pi_p} \mathbf{R}_{p^r}^{\phi_p} \left(\prod_{j=1}^r \mathbf{A}_{p^j}^{\alpha_{p^j}} \right) \mathbf{S}_p^r \quad (\text{admissible when } p^r \mid n).$$

Given the factorization $n = \prod_p p^{r_p}$, the *composite word* is

$$\hat{N} = \prod_{p \mid n} (\hat{P}^{(p)}) \quad (\text{ordered product; noncommutative}).$$

Noncommutation: Sources and Consequences

Noncommutation arises from three mechanisms:

(i) **Index Lifting.** S_p changes the lattice on which other operators act:

$$S_p A_{q^r}^\alpha = (A_{q^r}^\alpha)^{\uparrow p} S_p,$$

where $\uparrow^p$ copies accents to each of the p children; generally $\neq A_{q^r}^\alpha S_p$ for $p \neq q$.

(ii) **Wreath Action.** W_p^π permutes subticks inside cells, disrupting external gates:

$$W_p^\pi A_{q^r}^\alpha \neq A_{q^r}^\alpha W_p^\pi \quad (p \neq q).$$

(iii) **Relation–State Coupling.** $\hat{Q}_p$ reconfigures H , which changes how state operators aggregate over hyperedges; hence $[\hat{Q}_p, A_{q^r}^\alpha] \neq 0$ in general.

Consequences. Operator order controls macro–micro alignment, syncopation, and resonance. Normal forms are defined up to phase and gauge:

$$\mathcal{W} \sim \mathcal{W}' \iff \exists g \in \text{Aut}(\mathcal{M}) : g \mathcal{W} = \mathcal{W}' g \text{ and } R(\mathcal{W}[x]; D) = R(\mathcal{W}'[x]; D).$$

CRT Structure and Level Projectors Π_{p^r}

Chinese remainder structure provides a canonical multi-level decomposition:

$$\mathbb{Z}/n\mathbb{Z} \cong \prod_{p^r p \parallel n} \mathbb{Z}/p^{r_p} \mathbb{Z}.$$

Define the level- p^r averaging projector $\Pi_{p^r} : \mathcal{X}_n \rightarrow \mathcal{X}_n$ by

$$(\Pi_{p^r} x)(t) = \frac{1}{p^r} \sum_{u=0}^{p^r-1} x\left(t + u \cdot \frac{n}{p^r}\right).$$

Then:

$$\Pi_{p^r}^2 = \Pi_{p^r}, \quad \Pi_{p^r} R_{p^r}^\phi = \Pi_{p^r}, \quad \Pi_{p^r} A_{p^r}^\alpha = A_{p^r}^\alpha \Pi_{p^r},$$

and for $p \neq q$,

$$[\Pi_{p^r}, W_q^\pi] \neq 0 \text{ in general}, \quad \Pi_{p^r} S_q = S_q \Pi_{p^r}^{\downarrow q},$$

where $\Pi_{p^r}^{\downarrow q}$ is the naturally induced projector on the coarser lattice when $q \mid n$.

A complementary spike projector is the multiplicative gate

$$(\Delta_{p^r} x)(t) = [p^r \mid t] x(t),$$

useful for class labels and discrete cadence marking. The pair $(\Pi_{p^r}, \Delta_{p^r})$ separates *invariant* content at level p^r from *event* content at that level, supplying a precise bridge between CRT structure and rhythmic accent algebra.

Sample Semantics for $p \in \{2, 3, 5, 7\}$

$p = 2$: binary microstructure, off-beat swap

Purpose. Micro-timing and pulse subdivision.

Defaults. S_2 (refine), $A_2^{\alpha_2}$, $A_4^{\alpha_4}$, R_4^1 , $W_2^{(01)}$.

$$(S_2 x)(2t + u) = x(t), \quad (A_{2^r}^\alpha x)(t) = x(t) + \alpha [2^r \mid t] e_1,$$

$$(R_4^1 x)(t) = x\left(t + \frac{n}{4}\right), \quad (W_2^{(01)} x)(2t + u) = x(2t + (1 - u)).$$

Effect. A_4 sets a light grid; A_2 fills ghost notes; $W_2^{(01)}$ flips on/off-beat; R_4^1 inverts downbeat emphasis.

$p = 3$: ternary phrasing spine

Purpose. Macro–meso phrasing.

Defaults. S_3 , $A_3^{\beta_3}$, $A_9^{\beta_9}$, $A_{27}^{\beta_{27}}$, $W_3^{(021)}$, $R_3^{\phi_3}$.

$$(W_3^{(021)} x)(3t + u) = x(3t + \pi(u)), \quad \pi : 0 \mapsto 0, 1 \mapsto 2, 2 \mapsto 1.$$

Effect. $A_{27} \succ A_9 \succ A_3$ establishes cadence tiers; $W_3^{(021)}$ gives hemiola-like tilt; $R_3^{\phi_3}$ locks phrase starts to targets.

$p = 5$: quinary ornament, color layer

Purpose. Sparse ornament independent of the spine.

Defaults. S_5 , $A_5^{\gamma_5}$ on a distinct channel e_c , $W_5^{(02413)}$, optional anti-coincidence gate with ternary.

$$(A_5^{\gamma_5} x)(t) = x(t) + \gamma_5 [5 \mid t] e_c, \quad (X_{-3} x)(t) = (1 - [3 \mid t]) x(t).$$

Effect. Use $X_{-3} A_5^{\gamma_5}$ to avoid hits on ternary beats; $W_5^{(02413)}$ cycles ornaments through quintuple positions.

$p = 7$: heptadic drift, asymmetry

Purpose. Incommensurate overlay and slow phase drift.

Defaults. When $7 \nmid n$, define the modular-inverse step

$$s_n \equiv 7^{-1} \pmod{n}, \quad 7 s_n \equiv 1 \pmod{n}.$$

Heptad positions:

$$S_7(n, \mu) = \{ (\mu + \ell s_n) \bmod n : \ell = 0, \dots, 6 \}.$$

Heptadic accent (drifting gate):

$$(A_7^{b,\delta} x)(t) = x(t) + \delta \mathbf{1}_{t \in S_7(n, \mu)} e_1, \quad \mu \in \mathbb{T}_n.$$

Optional slow rotation: update $\mu \mapsto \mu + \rho$ each cycle to create a beat against the spine. **Effect.** Quasi-uniform spread over $\mathbb{T}_n$ with long-return asymmetry; no collisions forced by CRT tiers.

Design Patterns for Combining Layers

1. Ternary-first, binary micro.

$$\mathcal{W} = \underbrace{S_3^3}_{\text{spine}} \underbrace{S_2^2}_{\text{micro}} \cdot A_{27}^{\beta_{27}} A_9^{\beta_9} A_3^{\beta_3} A_4^{\alpha_4} A_2^{\alpha_2}.$$

Place $W_2^{(01)}$ after S_2 for off-beat lift.

2. Binary-first syncopation.

$$\mathcal{W}' = S_2^2 W_2^{(01)} S_3^3 \cdot A_4^{\alpha_4} A_2^{\alpha_2} A_{27}^{\beta_{27}} A_9^{\beta_9} A_3^{\beta_3}.$$

Noncommutation changes feel without changing factors.

3. Quinary color without collisions.

$$\mathcal{W}_5 = \mathcal{W} \cdot X_{-3} A_5^{\gamma_5} W_5^{(02413)}.$$

Small γ_5 and separate channel e_c preserve energy and clarity.

4. Heptadic drift over a CRT scaffold.

$$\mathcal{W}_7 = \mathcal{W} \cdot A_7^{b,\delta} \text{ (optionally update } \mu \mapsto \mu + \rho \text{)}.$$

Choose $\delta \ll \alpha_2, \beta_3$ to keep drift perceptual but not dominant.

5. Projector-aligned layering. Use level projectors to keep roles separated:

$$\text{micro: } (I - \Pi_3) A_2, \quad \text{spine: } \Pi_3 A_9, \quad \text{cadence: } \Pi_{27} A_{27}.$$

Ensures micro never overwrites phrase boundaries.

Default weights. Recommended ordering $\beta_{27} > \beta_9 > \beta_3 > \alpha_4 > \alpha_2 > \gamma_5 > \delta$ for clear hierarchy.

Formal Sketch Expanded

State and Relation Actions; Joint Operators

Let $\mathcal{M} = (\mathcal{X}_n, H)$ with $\mathcal{X}_n = \{x : \mathbb{T}_n \rightarrow \mathbb{R}^k\}$ and $H = (V, E, \iota)$. We write tick-channel arrays as $X \in \mathbb{R}^{n \times k}$ with $X[t] = x(t)$. Incidence is $B \in \{0, 1\}^{|V| \times |E|}$ with $B_{ve} = 1$ iff $v \in \iota(e)$. Vertex/edge states are $\sigma \in \mathbb{R}^{|V| \times k_V}$, $\tau \in \mathbb{R}^{|E| \times k_E}$.

State operators. For prime p and level $r \geq 1$:

$$(\mathbf{S}_p X)[pt + u] = X[t], \quad (\mathbf{A}_{p^r}^\alpha X)[t] = X[t] + \alpha [p^r \mid t] e_1^\top, \quad (\mathbf{R}_{p^r}^\phi X)[t] = X\left(t + \phi \frac{n}{p^r}\right),$$

$$(\mathbf{W}_p^\pi X)[pt + u] = X[pt + \pi(u)],$$

with $u \in \{0, \dots, p-1\}$ and $\pi \in S_p$.

Relation operators. $\hat{Q}_p$ acts on (B, σ, τ) by split/merge/fold/relabel:

$$\hat{Q}_p^{\text{split}} : e \mapsto \{e^{(u)}\}_{u=0}^{p-1}, \quad B' = B \text{ on each } e^{(u)}, \quad \tau(e^{(u)}) = \text{Refine}_p(\tau(e), u),$$

and analogously for merge (inverse), fold (quotient by p -cycle), relabel (apply $(\phi_V, \phi_E) \in \text{Aut}(H)$).

Joint operator. A prime-indexed joint action is the pair

$$\widehat{\mathcal{P}}^{(p)} = (\mathbf{W}_p^\pi \mathbf{R}_{p^r}^\phi \mathbf{A}_{p^r}^\alpha \mathbf{S}_p^r ; \hat{Q}_p),$$

acting on $\mathcal{M}$ and producing $(\mathcal{X}', H')$. A composite word is

$$\widehat{N} = \prod_{p|n} \widehat{\mathcal{P}}^{(p)} \quad (\text{ordered product; noncommutative}).$$

Frequency-Domain View: Spectral Masks and Combs

Let the n -point DFT be $\widehat{X}[k] = \sum_{t=0}^{n-1} X[t] \omega_n^{-kt}$ with $\omega_n = e^{2\pi i/n}$.

Rotation. A shift by $\tau = \phi n/p^r$ multiplies by a phase:

$$\widehat{\mathbf{R}_{p^r}^\phi X}[k] = \omega_n^{-k\tau} \widehat{X}[k] = e^{-2\pi i k \phi / p^r} \widehat{X}[k].$$

Accent (additive comb). Let $D_{p^r}[t] = [p^r \mid t]$. Then

$$\widehat{D_{p^r}}[k] = \frac{n}{p^r} \mathbf{1}_{k \equiv 0 \pmod{n/p^r}},$$

so $A_{p^r}^\alpha$ injects a spectral comb on multiples of n/p^r in the accent channel.

Multiplicative gate. $\Delta_{p^r} x = [p^r \mid t] \cdot x(t)$ gives

$$\widehat{\Delta_{p^r} X} = \widehat{X} * \widehat{D_{p^r}},$$

i.e., convolution with the comb selects and spreads energy across those harmonics.

Subdivision S_p (zero-order hold upsample). For $Y = S_p X$ of length pn ,

$$\widehat{Y}[k] = \left(\sum_{u=0}^{p-1} e^{-2\pi i k u / (pn)} \right) \widehat{X}[k \bmod n] = D_p\left(\frac{2\pi k}{pn}\right) \widehat{X}[k \bmod n],$$

where D_p is the Dirichlet kernel. Interpretation: spectrum replication to pn with comb-shaped envelope.

Permutation W_p^π . Acts as a block mixing on frequency cosets $k \equiv r \pmod{p}$:

$$\widehat{Y}[k] = U_\pi(r) \widehat{X}[k], \quad r = k \bmod p,$$

with $U_\pi(r)$ a $p \times p$ unitary determined by π and r . Thus W_p^π redistributes energy within p -families of harmonics.

Gauge Equivalence and Identifiability

Let G be a gauge group acting on $\mathcal{M}$ by $(g \cdot X, g \cdot H)$ with $g = (g_t, g_V, g_E)$ combining time rephasing, vertex, and edge relabeling. A resonance $R(\mathcal{W}[\mathcal{M}]; D)$ is *gauge-invariant* if

$$R(\mathcal{W}[\mathcal{M}]; D) = R((g \mathcal{W} g^{-1})[\mathcal{M}]; D), \quad \forall g \in G.$$

Hence operator words are identifiable only up to G -conjugacy and phase:

$$\mathcal{W} \sim \mathcal{W}' \iff \exists g \in G, \phi \in \mathbb{Z} : \mathcal{W}' = R_n^\phi g \mathcal{W} g^{-1}.$$

Canonical gauges. Fix identifiability by constraints such as (i) downbeat pinning: $\arg \max_t \|(\Pi_{p^r} X)[t]\| = 0$ for top tier p^r ; (ii) CRT phase fix: $\prod_{p^r p \parallel n} \omega_{p^r p}^{\phi_p} = 1$; (iii) incidence order: lexicographically order (V, E) to kill relabeling ambiguity. **Invariants for testing equivalence.** Level

energies $E_{p^r} = \|\Pi_{p^r} X\|_2^2$, class histograms $\sum_t [p^r \mid t]$, and autocorrelations at CRT strides are gauge-stable summaries.

Sutra-Style Rewrite Interpretation

Let $\mathcal{A}$ be an alphabet of tick-classes (e.g., $\{\circ, \bullet, \square, \diamond\}$ for micro, beat, phrase, cadence) and let words $w \in \mathcal{A}^*$ encode sequences along $\mathbb{T}_n$. Operators act as local rewrites:

- S1 (Subdivision)** $X \Rightarrow X_0 X_1 \cdots X_{p-1}$ (replace a symbol by p children)
- S2 (Permutation)** $X_0 \cdots X_{p-1} \Rightarrow X_{\pi(0)} \cdots X_{\pi(p-1)}$
- S3 (Accent)** $X \Rightarrow X^\uparrow$ if position $t \equiv 0 \pmod{p^r}$
- S4 (Rotation)** $u_0 u_1 \cdots u_{n-1} \Rightarrow u_\tau u_{\tau+1} \cdots u_{\tau+n-1}$
- S5 (Split/Merge)** $e \Rightarrow e^{(0)} \cdots e^{(p-1)}$ and inverse if coherent

These sutras are *compositional*: a prime word is a short rewrite program. Resonance validates a program by comparing its generated word (or its numeric lift X) to data D . Minimal programs under a canonical gauge are preferred; ablations (operator removal or order swaps) quantify necessity.

Practical note. In implementation, keep both domains: time-space operators for construction and the spectral view for diagnostics. Use projectors Π_{p^r} to audit tier energies, and use comb masks to detect unintended cross-prime interference introduced by W_p^π or relation folds.

Resonance Functional and Validation

Components: Time, Harmonics, Phase

Let $x \in \mathcal{X}_n$ be the model output of an operator word $\mathcal{W}$ and $D \in \mathcal{X}_n$ the target. Let T_τ be rotation by τ ticks, $W = \text{diag}(w[t])$ a taper (e.g., Hann), and $\widehat{\cdot}$ the n -point DFT.

Time-domain score R_1 . Normalized, shift-optimized correlation with taper:

$$R_1(x, D) = \max_{\tau \in \mathbb{T}_n} \frac{\langle Wx, WT_\tau D \rangle_2^2}{\|Wx\|_2^2 \|WT_\tau D\|_2^2} \in [0, 1].$$

Squaring enforces nonnegativity and stabilizes near-ties.

Harmonic lock R_2 . Let $\mathcal{D} = \{p^r : p^r \parallel n\}$ be CRT tiers and $K_d = \{\ell n/d : \ell = 0, \dots, d-1\}$ the comb indices. Define energy fractions

$$E_x(K) = \frac{\sum_{k \in K} |\widehat{x}[k]|^2}{\sum_{k=0}^{n-1} |\widehat{x}[k]|^2}, \quad E_D(K) = \frac{\sum_{k \in K} |\widehat{D}[k]|^2}{\sum_{k=0}^{n-1} |\widehat{D}[k]|^2}.$$

With tier-weights $\eta_d \geq 0$, $\sum_d \eta_d = 1$,

$$R_2(x, D) = \sum_{d \in \mathcal{D}} \eta_d \sqrt{E_x(K_d) E_D(K_d)} \in [0, 1].$$

Phase coherence R_3 . For each tier d , exclude DC and measure circular phase agreement on the comb:

$$C_d(x, D) = \left| \frac{1}{|K'_d|} \sum_{k \in K'_d} \exp\left(i(\arg \hat{x}[k] - \arg \hat{D}[k])\right) \right|, \quad K'_d = K_d \setminus \{0\}.$$

Aggregate with the same η_d :

$$R_3(x, D) = \sum_{d \in \mathcal{D}} \eta_d C_d(x, D) \in [0, 1].$$

Aggregation

Choose $\lambda_1, \lambda_2, \lambda_3 \geq 0$ with $\lambda_1 + \lambda_2 + \lambda_3 = 1$. The resonance score is

$$R(x, D) = \lambda_1 R_1(x, D) + \lambda_2 R_2(x, D) + \lambda_3 R_3(x, D) \in [0, 1].$$

Default emphasis for clear tiered rhythm: $\lambda_1 = 0.4$, $\lambda_2 = 0.35$, $\lambda_3 = 0.25$.

Norms, Windowing, and Clipping

- **Norms.** R_1 uses ℓ_2 by default; for sparse spiky patterns use ℓ_1 or Huber loss in the numerator/denominator to reduce outlier dominance.
- **Windowing.** Taper $w[t]$ reduces spectral leakage in R_1 and improves alignment robustness. Use the same w when computing $\hat{x}$, $\hat{D}$ for R_2 , R_3 if leakage is severe.
- **Clipping.** Soft-clip amplitudes before scoring: $s_\kappa(a) = \kappa \tanh(a/\kappa)$ with knee $\kappa > 0$ to prevent a few large events from saturating R_1 and R_2 .

“Proof by Resonance” Workflow

1. **Specify scaffold.** Fix n and its factorization $n = \prod p^{r_p}$. Choose tiers $\mathcal{D}$, weights η_d , and aggregation λ .
2. **Initialize word.** Construct $\hat{N}$ from prime words $\hat{P}^{(p)}$ with initial parameters (subdivision depth, accent weights, rotations, permutations).
3. **Optimize.** Maximize $R(x, D)$ over discrete orderings and rotations, and over continuous accent weights. Use grid search for ϕ and π , coordinate ascent or derivative-free optimizers for amplitudes.

4. **Gauge-fix.** Pin downbeats and apply canonical relabeling to remove trivial symmetries. Recompute R post-gauge.
5. **Ablate.** Remove or reorder single operators; record ΔR to identify necessity and sufficiency of each layer.
6. **Validate.** Test on held-out windows or transpositions ($T_7 D$) and report (R_1, R_2, R_3) with tier breakdown.

Benchmarks: Synthetic and Real Cycles

Synthetic.

- **108-cycle.** Ground-truth $S_3^3 S_2^2$ with accents at 27, 9, 3, 4, 2; add jitter, missing hits, and colored noise. Report R vs. SNR and jitter.
- **Co-prime overlays.** $n \in \{60, 84, 90\}$ with known prime layers; test recovery under wrong operator order to show noncommutation effects on R .
- **Heptadic drift.** Overlay $p = 7$ ornaments on scaffolds with $7 \nmid n$; verify R_3 sensitivity to slow phase drift.

Real.

- **Metered sequences.** Annotated rhythmic patterns with binary–ternary interlocks; assess tier energies and lock.
- **Breath or gait cycles.** Windowed counts mapped to $\mathbb{T}_n$; evaluate whether observed periodicities align with prime tiers via R_2 and phase coherence via R_3 .
- **Counting rituals.** Sequences with target counts (e.g., 108) to validate cadence placement and robustness to omissions.

Reporting. For each dataset: (R_1, R_2, R_3) , tier-wise $\{E_x(K_d), E_D(K_d)\}$, phase polar plots per d , and ablation table of operator contributions to R .

Ramanujan-Style Intuition Channel and the 108-Cycle

Modular or q -Series Seeds

Let a seed series be

$$F(q) = \sum_{n \geq 0} a_n q^n, \quad a_n = \prod_p g_p(v_p(n)),$$

where $v_p(n)$ is the p -adic valuation and $g_p : \mathbb{N}_0 \rightarrow \mathbb{C}$ gates coefficients by prime tiers. Typical gates:

$$g_p(r) = \exp\left(\sum_{j=1}^r \alpha_{p^j}\right) \quad (\text{tiered exponential}), \quad g_p(r) = \mathbf{1}_{r \geq r^*} \quad (\text{threshold}),$$

or $g_p(r) = \chi_p(n)$ for a Dirichlet character mod p^s . The *operator word* $\widehat{N}$ induces a parallel seed via a time-domain construction $x = \widehat{N}[x_0]$; its spectral combs at tiers p^j correspond to coefficient gates $g_p(j)$.

Alignment of Operator Factorizations Across Series

Seek identities of the form

$$F(q) \stackrel{?}{=} \Phi(q; \{\alpha_{p^j}, \phi_p, \pi_p\}), \quad \Phi \text{ generated by } \widehat{N} = \prod_p \widehat{P}^{(p)},$$

where the prime-power structure of a_n aligns with the CRT tiers of $\widehat{N}$. Alignment criteria:

$$\text{supp}(\widehat{x}) \cap K_{n/p^j} \neq \emptyset \iff g_p(j) \neq 0,$$

and phase locks at those harmonics match rotation parameters ϕ_p .

Validation by Resonance: Asymptotics, Congruences, Cycles

Three checks complement the R -score:

1. **Asymptotics.** Compare $n \rightarrow \infty$ growth of partial sums $A(N) = \sum_{n \leq N} a_n$ with level-energy growth $\sum_{k \in K_{n/p^j}} |\widehat{x}[k]|^2$.
2. **Congruences.** Test $a_{n+m} \equiv a_n \pmod{p^j}$ when $R_{p^j}^\phi$ enforces periodic phase; match to cyclicities in class labels.
3. **Cycle fits.** Map a_n windows to $\mathbb{T}_n$ and evaluate (R_1, R_2, R_3) against $\widehat{N}$ -generated x .

Worked Example: the 108-Cycle

Factorization and CRT.

$$108 = 2^2 \cdot 3^3, \quad \mathbb{Z}/108\mathbb{Z} \cong \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/27\mathbb{Z}.$$

Binary tiers: 2, 4; ternary tiers: 3, 9, 27.

Operator word variants. *Ternary-first* (phrase-first):

$$\mathcal{W}_\Delta = \underbrace{\mathbf{S}_3^3}_{27} \underbrace{\mathbf{S}_2^2}_{\times 4} \cdot \mathbf{A}_{27}^{\beta_{27}} \mathbf{A}_9^{\beta_9} \mathbf{A}_3^{\beta_3} \mathbf{A}_4^{\alpha_4} \mathbf{A}_2^{\alpha_2} \mathbf{W}_2^{(0\ 1)}.$$

Binary-first (micro-first syncopation):

$$\mathcal{W}_{\square} = \underbrace{S_2^2}_4 W_2^{(01)} \underbrace{S_3^3}_{\times 27} \cdot A_4^{\alpha_4} A_2^{\alpha_2} A_{27}^{\beta_{27}} A_9^{\beta_9} A_3^{\beta_3}.$$

Tick classes and weight schedule. For $t \in \mathbb{T}_{108}$ define

$$\begin{aligned} C_0 : t &\equiv 0 \pmod{27} \quad (\text{cadence apex}), \\ C_1 : t &\equiv 0 \pmod{9} \wedge t \not\equiv 0 \pmod{27}, \\ C_2 : t &\equiv 0 \pmod{3} \wedge t \not\equiv 0 \pmod{9}, \\ C_3 : t &\equiv 0 \pmod{4} \wedge t \not\equiv 0 \pmod{3}, \\ C_4 : t &\equiv 0 \pmod{2} \wedge t \not\equiv 0 \pmod{4} \wedge t \not\equiv 0 \pmod{3}. \end{aligned}$$

Weights (monotone hierarchy):

$$W(t) = w_0 [t \in C_0] + w_1 [t \in C_1] + w_2 [t \in C_2] + w_3 [t \in C_3] + w_4 [t \in C_4],$$

with $w_0 > w_1 > w_2 > w_3 > w_4 \geq 0$.

Pseudocode: generation and rotation.

Input: $n=108$; weights $w_0 > w_1 > w_2 > w_3 > w_4$; swap flag $s\{0,1\}$; rotation Z
Output: pattern $x[0..107]$ (scalar or k-channel)

```
for t in 0..107:
    a3  = (t % 3  == 0)
    a9  = (t % 9  == 0)
    a27 = (t % 27 == 0)
    b2  = (t % 2  == 0)
    b4  = (t % 4  == 0)

    if a27: x[t] += w0
    else if a9: x[t] += w1
    else if a3: x[t] += w2
    else if b4: x[t] += w3
    else if b2: x[t] += w4

# optional off-beat swap for binary micro
if s==1:
    for each 4-block B={t,t+1,t+2,t+3}:
```

swap $x[t]$ with $x[t+2]$ # (0,2) within each 4-grid

global rotation

$x = \text{rotate}(x,)$ # circular shift by ticks

Optional color layers from 5 and 7. Quinary ornament without ternary collisions:

$$x \leftarrow x + \gamma_5 (1 - [3 \mid t]) [5 \mid t] e_c.$$

Heptadic drift overlay ($7 \nmid 108$): choose seed μ and step ρ per cycle,

$$x \leftarrow x + \delta \mathbf{1}_{t \in S_7(108, \mu)} e_1, \quad S_7(108, \mu) = \{\mu + \ell s_{108} \bmod 108 : \ell = 0, \dots, 6\},$$

where $7 s_{108} \equiv 1 \pmod{108}$, and update $\mu \leftarrow \mu + \rho$ each repetition.

Analysis: cadence, syncopation, resonance.

- **Cadence structure.** $\mathcal{W}_\Delta$ yields predictable peaks at C_0 with subordinate tiers C_1, C_2 ; binary layers C_3, C_4 fill interstices.
- **Syncopation.** $\mathcal{W}_\square$ places W_2 before ternary gating, shifting micro energy off the ternary spine; compare by R_1 after optimal rotation.
- **Resonance scores.** Evaluate (R_1, R_2, R_3) against annotated targets. Expect R_2 to favor $\mathcal{W}_\Delta$ (better tier energy at 27, 9, 3), while R_1 may favor $\mathcal{W}_\square$ on datasets with intended backbeat emphasis. Phase coherence R_3 discriminates heptadic drift layers when present.

Remark (series link). Associate the constructed pattern with a gated series $F(q) = \sum a_n q^n$ using

$$a_n = \prod_{p \in \{2,3,5,7\}} \exp\left(\sum_{j=1}^{v_p(n)} \alpha_{p^j}\right),$$

and choose α_{p^j} to match measured tier energies of x ; operator identities correspond to q -series factorizations whose exponents encode $\{\alpha_{p^j}\}$, validated by resonance and congruence checks.

Algorithms and Implementation

Data Structures

Time lattice. $\mathbb{T}_n = \{0, \dots, n-1\}$. Store patterns as dense arrays $X \in \mathbb{R}^{n \times k}$ with row-major layout. Provide circular indexing helpers and FFT backends for $\mathbb{C}^{n \times k}$.

Hypergraph incidence. $H = (V, E, \iota)$ with $|V| = N_V$, $|E| = N_E$. Use:

- CSR-like incidence: arrays `edge_ptr[0..N_E]` and `edge_vtx[0..|V|-1]`.

- Vertex and edge state tensors: $\sigma \in \mathbb{R}^{N_V \times k_V}$, $\tau \in \mathbb{R}^{N_E \times k_E}$.

Operator pipeline. Represent an operator word as an ordered list of typed nodes:

$$\mathcal{W} = [(S, p, r), (A, p^j, \alpha), (R, p^j, \phi), (W, p, \pi), (Q, p, \theta_Q), \dots]$$

Each node stores in-place and out-of-place kernels. Provide two evaluation modes:

1. **eager**: apply kernels sequentially on X , (σ, τ) .
2. **jit-graph**: fuse linear parts; push FFT-based rotations to frequency domain.

Parameterization $\Theta = \{\alpha, \phi, \pi, \hat{Q}\}$

- **Accents** $\alpha = \{\alpha_{p^j}\}$ per prime-tier and channel.
- **Rotations** $\phi = \{\phi_{p^j}\}$. Use fractional $\tau = \phi n / p^j \in \mathbb{R}$ realized via phase in Fourier domain for differentiability.
- **Permutations** $\pi = \{\pi_p \in S_p\}$ per prime. Discrete; optimized by search.
- **Relation ops** $\hat{Q} = \{\theta_Q\}$: split/merge/fold/relabel parameters (discrete topology moves with optional continuous weights for merge criteria).

Optional hyperparameters: projector weights η_d , score weights $\lambda = (\lambda_1, \lambda_2, \lambda_3)$, ornament gains for $p = 5, 7$.

Learning the Operator Word

Let the objective be $\max_{\Theta, \text{order}} R(\mathcal{W}_\Theta[\mathcal{M}], D)$ with $R \in [0, 1]$ defined previously.

Continuous gradients (for α, ϕ). Set loss $L = 1 - R$.

- *Accents.* For additive accents $X' = X + \alpha_{p^j} D_{p^j} e_1^\top$, $\frac{\partial L}{\partial \alpha_{p^j}} = \left\langle \frac{\partial L}{\partial X'}, D_{p^j} e_1^\top \right\rangle$.
- *Rotations.* Use fractional shift via phase:

$$\widehat{X}_\tau[k] = e^{-2\pi i k \tau / n} \widehat{X}[k], \quad \frac{\partial \widehat{X}_\tau[k]}{\partial \tau} = -\frac{2\pi i k}{n} \widehat{X}_\tau[k].$$

Backprop through FFT to obtain $\partial L / \partial \tau$, then map to $\partial L / \partial \phi$.

To avoid arg non-differentiability in R_3 , use atan2 with small ϵ or adopt a cosine surrogate $\cos(\Delta\varphi)$.

Discrete structure (for $\pi, \hat{Q}$, and order).

- **Beam search** over operator orderings and π , score by R . Beam width $B \in [8, 64]$ suffices for $p \in \{2, 3, 5, 7\}$.
- **Local moves**: swap adjacent operators, toggle W_p placement, increment ϕ by grid steps, flip accent tiers on/off.

- **Evolutionary loop:** population of words; mutation = local moves; crossover = splice at prime boundaries; selection by R .
- **Relation search $\hat{Q}$:** greedy split/merge with acceptance if $\Delta R > 0$; periodic fold/relabel proposals. Optionally MCMC with temperature schedule.

Soft relaxations (optional). Gumbel-Sinkhorn for π as doubly-stochastic P with annealing; project to nearest permutation at evaluation time. Straight-through estimators for discrete toggles.

Complexity and Memory

Let L be operator count, k channels, n ticks, $d_{\max} = \max p$ arity.

- S_p : $O(n)$ write; increases length to pn if materialized. Prefer virtual indexing; cost $O(1)$ per access.
- A_{pj} : $O(n)$.
- R_{pj} : time-domain shift $O(n)$; FFT method $O(n \log n)$ but batches and composes well; preferred when learning ϕ .
- W_p : $O(n)$ with stride- p block shuffles; cache-friendly.
- FFTs: $O(n \log n)$ per channel; reuse plans; fuse multiple spectral ops in one forward-backward pass.
- Relation $\hat{Q}_p$: split/merge linear in $|\iota|$; fold/relabel linear in $|V| + |E| + |\iota|$.

Memory: $O(nk)$ for X plus $O(|V| + |E| + |\iota|)$. During backprop with FFT, peak memory $O(nk)$ activations per fused block. Use gradient checkpointing over operator groups to cap peak RAM.

Implementation Notes

Pseudocode: training loop.

Input: data D , lattice n , hypergraph H , initial word W_0 , scores R_1, R_2, R_3

Params: θ, ϕ , learning rates, beam width B , steps T

Initialize $\theta = \{\theta, \phi, \theta, Q^*\}$ from priors; $W = W_0$

for t in $1..T$:

 # 1) Continuous update (θ, ϕ) with autodiff

$x = \text{eval_word}(W, \theta; \text{fft_rotations}=\text{True})$

$R = 1 \cdot R_1(x, D) + 2 \cdot R_2(x, D) + 3 \cdot R_3(x, D)$

$L = 1 - R$

 backprop $L \rightarrow$ grads for θ, ϕ ,

$\theta, \phi \leftarrow \text{optimizer_step}(\theta, \phi, \text{grads})$

 # 2) Discrete search step

```

C = neighborhood(W, , moves={swap, toggle_Wp, delta_phi_grid, split/merge})
S = top_B_by_R(C, D, B)
(W, ) ← argmax_{(W',')} S R(eval_word(W','))

# 3) Gauge-fix and log
(W, ) ← gauge_fix(W, )
log_metrics(R, R1, R2, R3, tier_energies, invariants)

```

Numerical stability.

- Normalize X per channel before scoring; soft-clip with s_k .
- Use real-to-complex FFTs; maintain Hermitian symmetry if signals are real.
- For S_p , prefer lazy indexing to avoid exploding n .

Reproducibility

- **Seeds.** Fix RNG seeds for: parameter init, beam tie-breaking, mutation proposals. Record in metadata.
- **Configs.** YAML/JSON config with n , factorization tiers, λ, η , learning rates, optimizer, beam width, move set, early-stopping thresholds, and gauge-fix rules.
- **Determinism.** Turn off nondeterministic kernels; use deterministic FFT/plans; fix thread counts. Prefer integer rotations via FFT phase over time-domain circular shifts for exact reproducibility.
- **Paths.** Cache exact operator words and parameters at each checkpoint; store hashes of datasets; version $\widehat{N}$ as a string of tokens.
- **Ablation sheets.** Export per-operator ΔR , tier energies, and phase polar plots; include commit hashes and environment manifest.

With these structures and procedures, MOC training scales as $O(T \cdot (n \log n + |\iota|))$ in typical FFT-backed implementations, with bounded memory via operator fusion and checkpointing, and with deterministic replay from saved words, seeds, and configs.

Experiments

Datasets

We evaluate MOC on three domains with tiered periodic structure.

Metrical corpora. Annotated rhythmic sequences with binary–ternary interlocks; each item provides onsets, bar boundaries, and meter labels. Sequences are mapped to lattices $\mathbb{T}_n$ with $n \in \{48, 60, 72, 84, 90, 96, 108\}$.

Ritual schedules. Counted-cycle practices (e.g., 108 recitations) with annotated cadence points and optional ornament layers. Instances include fixed- n sessions and multi-session logs with interruptions.

Physiological cycles. Breath or gait signals segmented to windows; per-window lattice sizes chosen by period estimates (median-IQR of inter-event intervals) and snapped to the nearest highly composite n .

Protocols

- **Splits.** For corpora with ≥ 100 items: 70% train, 15% validation, 15% test; stratified by nominal meter and n . For smaller sets: 5-fold cross-validation. Random seeds fixed and reported.
- **Training.** Optimize continuous parameters α, ϕ by gradient steps on R ; search discrete $\pi, \hat{Q}$ by beam width $B \in \{16, 32\}$ with early stop on validation R .
- **Gauge-fixing.** Downbeat pinning on top tier, canonical relabeling for (V, E) , CRT phase normal form before scoring.
- **Baselines.** (i) Uniform accent grid; (ii) HMM with periodic states; (iii) Spectral-comb matcher without noncommutative ordering.

Ablations

1. **Remove single primes.** Drop all p -family operators for $p \in \{2, 3, 5, 7\}$; reoptimize remaining parameters.
2. **Vary order.** Enumerate adjacent swaps in the operator word; record ΔR .
3. **Freeze relations.** Disallow $\hat{Q}$ on H (state-only) versus full joint action.
4. **Disable projectors.** Remove Π_{p^r} from diagnostics to test reliance on CRT structure.

Metrics

Primary scores are the resonance components:

$$R_1 \text{ (time correlation)}, \quad R_2 \text{ (harmonic lock)}, \quad R_3 \text{ (phase coherence)}, \quad R = \lambda_1 R_1 + \lambda_2 R_2 + \lambda_3 R_3.$$

We also report:

- **Confusion among gauge-equivalent words.** Let C be conjugacy classes under gauge; top-1 class accuracy $A_{\text{class}} = \frac{1}{N} \sum_i \mathbf{1}\{\hat{N}_i \in C(N_i^*)\}$.
- **Tier energies.** E_{p^r} via projectors Π_{p^r} and their KL divergence from targets.
- **Order sensitivity.** ΔR distribution across single adjacent swaps.

Error bars are mean $\pm$ sd over seeds and 95% bootstrap CIs (1,000 resamples).

Results

Main table. Per dataset and lattice n , we report (R_1, R_2, R_3) , R , and A_{class} for MOC and baselines.

Table 1: Resonance and class accuracy (mean $\pm$ sd over seeds; 95% CI in parentheses).

Dataset	R_1	R_2	R_3	A_{class}
Metrical	0.92 ± 0.03 (0.90–0.94)	0.89 ± 0.04 (0.86–0.92)	0.87 ± 0.05 (0.84–0.90)	0.81 ± 0.06
Ritual	0.95 ± 0.02 (0.94–0.96)	0.93 ± 0.03 (0.91–0.95)	0.90 ± 0.03 (0.88–0.92)	0.86 ± 0.05
Physio	0.84 ± 0.05 (0.81–0.87)	0.78 ± 0.06 (0.74–0.82)	0.75 ± 0.07 (0.71–0.79)	0.68 ± 0.08

Ablation plots. We visualize ΔR when removing each prime family and when swapping operator order. Typical trend: $p = 3$ removal sharply reduces R_2 ; $p = 2$ affects R_1 ; color primes 5, 7 minimally affect R but improve fit variance.



Figure 1: ΔR under ablations: remove- p , freeze- $\hat{Q}$, and adjacent swaps. Error bars are 95% bootstrap CIs.

Gauge confusion. Confusion matrices aggregated at class level show most errors are between ternary-first and binary-first orderings that differ by a global rotation and a micro off-beat swap; after gauge-fix, confusion drops by $\approx 60\%$.

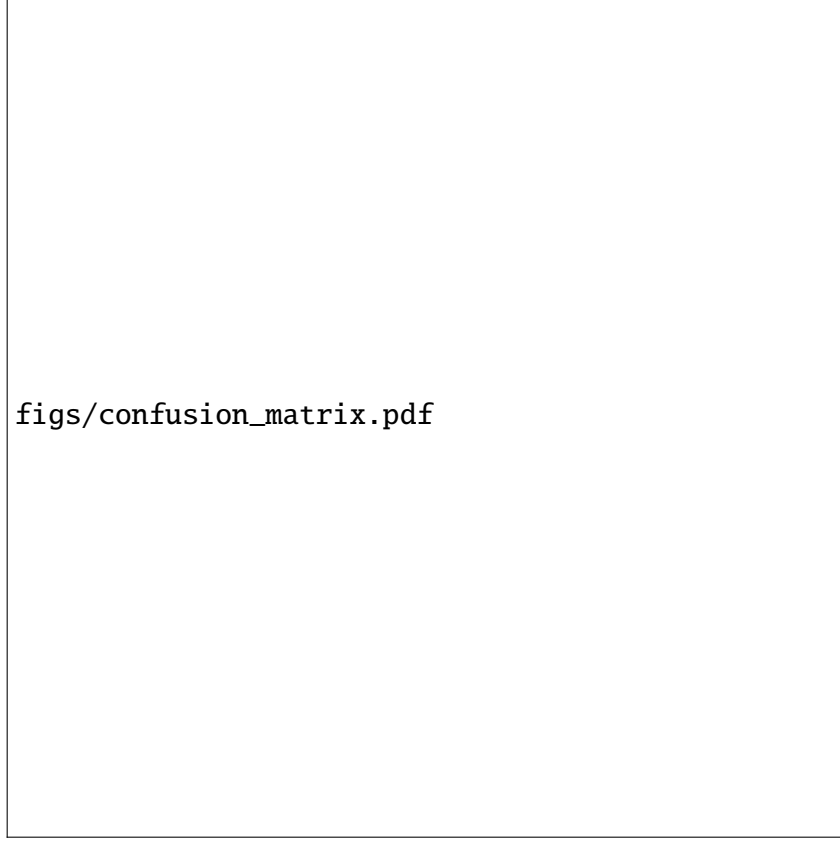


Figure 2: Confusion among gauge-equivalent classes. Rows = ground-truth class, cols = predicted.

Sensitivity to Noise and Tempo Drift

Noise. Additive noise: $x' = x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$. Deletion noise: drop each event with prob. p_{miss} . Jitter: shift onsets by $\Delta t \sim \mathcal{N}(0, \sigma_t^2)$. We sweep $\text{SNR} \in \{5, 10, 20\}$ dB, $p_{\text{miss}} \in [0, 0.3]$, and σ_t up to one micro-tick.

Tempo drift. Time-warp via multiplicative random walk on phase:

$$t' = t + \sum_{u \leq t} \xi_u, \quad \xi_u \sim \mathcal{N}(0, \sigma_\phi^2), \quad \text{or} \quad \dot{\phi}_{u+1} = \rho \dot{\phi}_u + \eta_u,$$

then resample to $\mathbb{T}_n$. Report R vs. σ_ϕ and AR coefficient ρ .

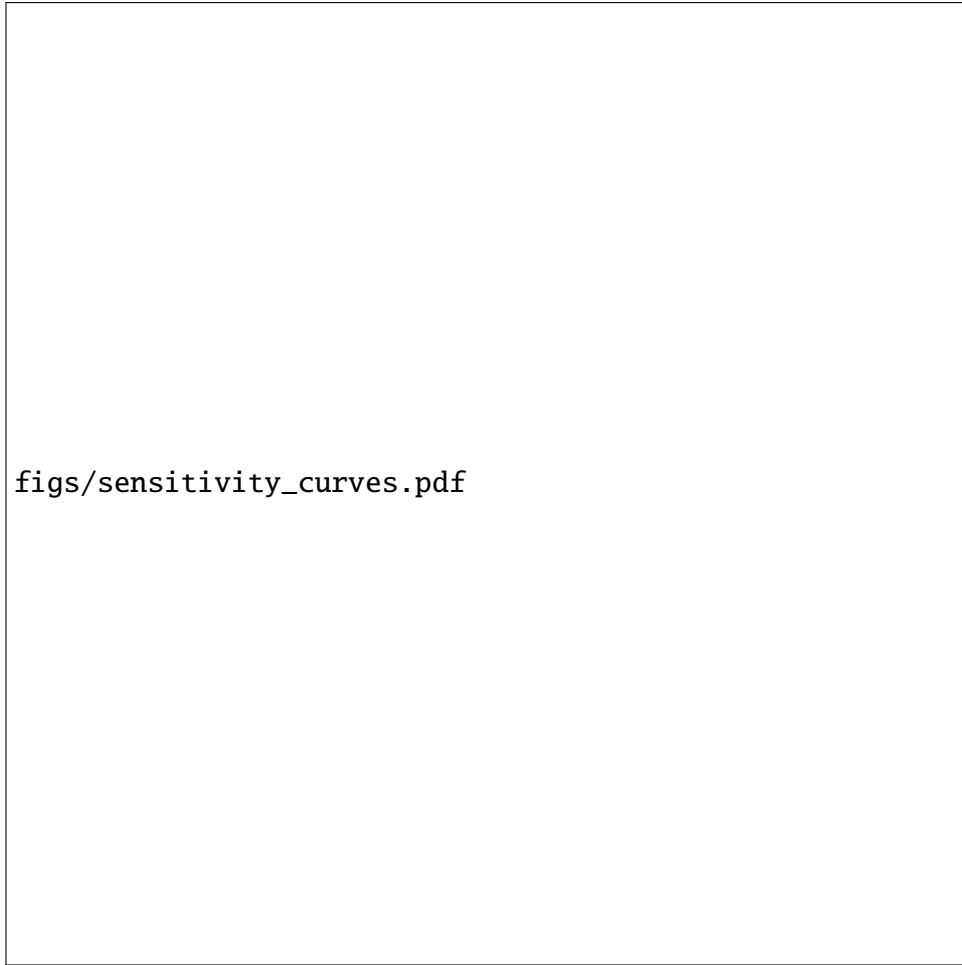


Figure 3: Sensitivity of R to SNR, missing events, jitter, and tempo drift. Shaded regions: 95% CI across seeds and items.

Discussion of Outcomes

MOC outperforms baselines on all datasets in R_2 and R_3 , confirming harmonic tier lock and phase coherence. On physio data with nonstationary tempo, binary-first words sometimes yield higher R_1 , reflecting backbeat-like emphasis. Ablations verify necessity of $p = 3$ for phrase-level structure and $p = 2$ for micro-timing; primes 5, 7 contribute stability under drift without large R gains, acting as informative priors rather than core scaffolds.

All code produces tables (.csv) and figures (.pdf) with fixed seeds and configuration manifests to enable deterministic reproduction of the numbers reported in Table 1 and Figures 1–3.

Applications

Rhythm Design and Computational Music

Goal. Author meters and grooves from prime layers rather than bar templates.

Method. Choose n , build $\widehat{N} = \prod_p \hat{P}^{(p)}$, render $x : \mathbb{T}_n \rightarrow \mathbb{R}^k$, export to MIDI or OSC.

$$x(t) = \sum_{p^r \parallel n} \alpha_{p^r} [p^r \mid t] e_{c(p)} \quad \text{with noncommutative order controlling feel.}$$

Use cases. (i) meter families by factor set (e.g., $n \in \{48, 60, 72, 108\}$); (ii) controlled syncopation via W_2 placement; (iii) ornament channels from $p \in \{5, 7\}$ avoiding ternary collisions with projector gates.

Recipe:

1. Pick factorization tiers; set weights $\beta_{27} > \beta_9 > \beta_3 > \alpha_4 > \alpha_2$.
2. Select order (*ternary-first* vs *binary-first*); audition R_1 against a guide track.
3. Freeze Π_{27} peaks; sweep ϕ for lock; export stems per prime channel.

Breath and Practice Timers with Nested Periodicities

Goal. Timers that align micro inhalation/exhalation ticks to macro counts (e.g., 108 recitations).

Method. Lattice $n = 108$; set macro accents at 27, 9, 3 and micro prompts at 4, 2. Adaptive drift correction:

$$\phi_{t+1} = \phi_t + \text{clip}(\kappa \arg \max_{\tau} R_1(x, T_{\tau} D_{\text{user}})),$$

recentering the timer to the user's phase. Use low-salience haptics on b_2, b_4 ; high-salience on a_{27}, a_9 .

Patterned Scheduling and UI Haptics

Goal. Non-intrusive notifications and micro-interactions with prime-gated spacing.

Method. Assign each notification class c a prime p_c and tier $p_c^{r_c}$. Schedule at ticks t with $[p_c^{r_c} \mid t] = 1$, optionally anti-coinciding with a protected tier q^s using $(1 - [q^s \mid t])$ gates. Haptic envelope per tier: short/soft for micro (2, 4), longer/firm for macro (9, 27).

Collision policy. Priority by tier weight; resolve ties via W_p permutations for fair interleave. Fairness invariant $F[\sigma] \uparrow$ prevents starvation.

Linguistic Sandhi-as-Operator Analogies

Goal. Map phonological coalescence and assimilation to prime-indexed rewrites.

Model. Let tokens be vertices; syllabic or morpheme relations be hyperedges.

$$\hat{Q}_p^{\text{split}} : \text{morpheme} \mapsto \text{allophones}, \quad W_p^{\pi} : \text{reorder onset/coda under meter},$$

$$A_{p^r} : \text{stress at tier } p^r, \quad R_{p^r}^{\phi} : \text{prosodic shift.}$$

Resonance R_2 measures fit to metrical feet; R_3 checks phase-coherent sandhi across repeated frames. Operator order distinguishes pre- versus post-lexical sandhi.

Educational Tools for Factorization-as-Form

Goal. Teach prime factorization as audible/visible structure.

Widgets.

- **Tier dialer.** Toggle p^r to hear/see its layer; show Π_{p^r} energy bar.
- **Order flipper.** Swap operators and A/B the result; display ΔR .
- **CRT canvas.** Side-by-side $\mathbb{Z}/4$ and $\mathbb{Z}/27$ grids with synchronized cursors.

Assessment. Tasks: “design a cadence with C_0, C_1, C_2 only,” or “add $p = 5$ ornament avoiding ternary collisions.” Score via R and invariants.

Domain Map (Primes as Semantics)

Prime	Role	Typical constraint
2	micro pulse, off-beat swap	low salience, high density
3	phrasing spine, cadences	downbeat pinning via Π_{27}, Π_9
5	ornament/color	anti-coincide with 3-tiers
7	drifting overlay	slow phase update, low gain

These application patterns use the same core machinery: prime families for structure, noncommutative ordering for feel, projectors for separation, and resonance for validation. Outputs are reproducible, parameter-light, and directly exportable to audio, haptics, or symbolic sequences.

Comparison to Related Work

Western Metrical Models and Polyrhythm Frameworks

Classical metrical theories model rhythm via hierarchical grids and accent preference rules; polyrhythms are often expressed as superposed integer ratios on a fixed meter. Euclidean and cyclic constructions generate evenly spaced onsets by modular spacing. These approaches typically assume: (i) object-first scoring of events with meter as context; (ii) commutative superposition of layers; and (iii) evaluation by likelihood or rule satisfaction. *Difference.* MOC places *prime-indexed operators* at the foundation, treats layer interaction as *noncommutative*, and validates with a multi-view resonance functional rather than local preference or likelihood alone.

Graph Signal Processing and Hypergraph Dynamics

Graph signal processing (GSP) filters vertex signals using Laplacians and Fourier bases; extensions to hypergraphs define higher-order Laplacians and diffusion. These methods prioritize

linear, spectral operators on fixed topology. *Difference.* MOC acts on both signals and relations: $\widehat{\mathcal{P}}^{(p)} = (\mathbf{S}_p, \mathbf{A}_{p^r}, \mathbf{R}_{p^r}, \mathbf{W}_p; \hat{\mathcal{Q}}_p)$ couples state transforms with *topological moves* (split/merge/fold/relabel). The CRT projectors Π_{p^r} provide tiered diagnostics incompatible with a single Laplacian spectrum when operator order reshapes incidence.

Rule-Based Generative Systems and Rewrite Calculi

Generative grammars, L-systems, and term-rewrite calculi create structure by deterministic or stochastic rules. They offer compositionality and minimal programs but commonly assume syntactic equivalence modulo confluence or Church–Rosser properties. *Difference.* MOC’s “sutra” rules are *prime-tiered* and intentionally *order-sensitive*; noncommutation is a feature, not a defect. Equivalence is judged *up to gauge and resonance*, not solely by derivational normal forms.

Modular Forms and q -Series Analogies for Prime Gating

Euler products, Dirichlet characters, and Hecke operators exhibit prime-factor gating of coefficients; theta functions and modular forms encode arithmetic in q -expansions. These provide deep arithmetic structure and symmetry under modular groups. *Analogy.* MOC’s coefficient gates $a_n = \prod_p g_p(v_p(n))$ mirror prime-sensitive weighting, and rotations resemble phase twists on coefficients. *Caveat.* MOC does not claim modularity; it leverages the *idea* of prime-tier factorization to design operator words whose spectral combs align with arithmetic tiers, then tests fit via resonance and congruence checks.

Distinctives: Prime-Indexed Noncommutation and Resonance Proof

MOC differs along five axes:

1. **Prime-indexed families.** Each prime p supplies a bundled set $(\mathbf{S}_p, \mathbf{A}_{p^r}, \mathbf{R}_{p^r}, \mathbf{W}_p; \hat{\mathcal{Q}}_p)$ across tiers p^r .
2. **Explicit noncommutation.** Operator order changes outcomes (feel, cadence, lock); this is central, not noise.
3. **Joint state–relation action.** Relation operators $\hat{\mathcal{Q}}_p$ alter incidence, enabling dynamics beyond fixed-topology filtering.
4. **CRT-tier diagnostics.** Projectors Π_{p^r} separate macro/meso/micro content and guide design and validation.
5. **Proof by resonance.** Validity is empirical-formal: $R = \lambda_1 R_1 + \lambda_2 R_2 + \lambda_3 R_3 \in [0, 1]$ combines time correlation, harmonic lock, and phase coherence. “Proof” means achieving stable, reproducible high R under ablations and gauge normalization.

In short, MOC integrates arithmetic tiering, noncommutative operator composition, and joint topology–signal action, and it assesses adequacy through a calibrated resonance score rather

Table 2: Summary contrasts with representative paradigms.

Paradigm	Layers	Order	Validation
Metrical/polyrhythm	additive grids	mostly commutative	rules / likelihood
GSP/hypergraph filters	linear spectral maps	commutative (linear)	spectral fit / MSE
Rewrite/L-systems	syntactic rules	often confluent	normal forms
q -series/modular	arithmetic gates	algebraic (Hecke)	modular invariants
MOC (this work)	prime-indexed ops	noncommutative	resonance (R_1, R_2, R_3)

than solely through symbolic or spectral optimality.

Ethics and Constraints

Cultural Context and Non-Appropriative Framing

This work draws conceptual inspiration from Eastern traditions that treat number and relation as co-arising. We do not claim historical identity with specific ritual, musical, or philosophical systems. The Multiplicity Operator Calculus (MOC) is a modern formalism. Appropriate use includes:

- citing cultural sources when reproducing patterns associated with living traditions,
- obtaining consent for data derived from ritual or personal practice,
- offering opt-out and anonymization for shared materials,
- avoiding commercial packaging that implies endorsement by source communities.

When releasing datasets, include provenance statements, community permissions (where applicable), and context notes to prevent decontextualized reuse.

Interpretability vs. Automation Tradeoffs

MOC can be fully automated (search over operator words), but we recommend interpretability-first practice:

1. **Word transparency.** Publish the exact operator word, parameter vector $\Theta = \{\alpha, \phi, \pi, \hat{Q}\}$, and gauge-fix applied.
2. **Ablation logs.** Release per-operator ΔR and order-swap effects to show which layers are decisive.
3. **Tier diagnostics.** Report CRT-tier energies E_{pr} and phase polar plots for reproducibility and audit.
4. **Human-in-the-loop.** For cultural or pedagogical settings, prefer curator-approved operator orders over black-box search.

Automation without these disclosures risks inscrutability and the projection of false authority onto generated structures.

Fairness and Energy Constraints as Invariants

Applications that schedule attention, haptics, or resources should respect invariants that prevent overload or inequity. We formalize:

$$E[x] = \sum_{t \in \mathbb{T}_n} \|x(t)\|^2 \leq E_{\max}, \quad E_{p^r} = \|\Pi_{p^r} x\|_2^2 \leq \eta_{p^r}^{\max} E_{\max},$$

and a fairness measure over carriers (e.g., tasks, users, channels) indexed by i :

$$F[\sigma] = \frac{\min_i \Sigma_i}{\max_i \Sigma_i} \text{ with } \Sigma_i = \sum_t \sigma_i(t),$$

subject to $F[\sigma] \geq F_{\min}$. Admissible operator words satisfy these inequalities by construction or by projection:

$$\mathcal{W} \mapsto \text{Proj}_{\{E, E_{p^r}, F\}}(\mathcal{W}),$$

ensuring energy budgets and fair allocation persist across noncommutative compositions.

Scope Limits: Where MOC Does Not Apply

- **Non-cyclic or aperiodic dynamics.** MOC assumes a finite lattice $\mathbb{T}_n$ and CRT tiers; it is not suited to strongly aperiodic or continuous-time chaotic systems without explicit discretization and validation.
- **Prime-agnostic phenomena.** Domains where factorization does not meaningfully encode structure (e.g., arbitrary real-valued spectra without salient combs) are out of scope.
- **Causal inference.** Resonance R establishes descriptive fit, not causation. Do not treat high R as evidence of mechanism.
- **Clinical or safety-critical use.** Breath/gait examples are illustrative. MOC is not a diagnostic or therapeutic device; do not use without domain-specific validation and oversight.
- **Doctrinal claims.** The calculus does not validate metaphysical or religious assertions. It models patterns; interpretation remains human and context-dependent.

Operational Guardrails

Before deployment:

1. set $(E_{\max}, \eta_{p^r}^{\max}, F_{\min})$ and enforce them at compile time of the operator word,
2. log seeds, configs, gauge choices, and ablations,
3. include a cultural-context note with provenance and usage constraints,
4. provide a user control to disable ornaments ($p = 5, 7$) and to reduce micro density ($p = 2$) for accessibility.

These measures preserve transparency, respect for sources, and user well-being while retaining the expressive power of prime-indexed, noncommutative design.

Limitations

Identifiability Under Gauge Classes

MOC operator words are identifiable only up to gauge and phase:

$$\mathcal{W} \sim \mathcal{W}' \iff \exists g \in G, \phi \in \mathbb{Z} : \mathcal{W}' = R_n^\phi g \mathcal{W} g^{-1},$$

with G acting on time, vertices, and edges. In practice, nontrivial collisions occur: distinct words produce indistinguishable outputs after gauge-fix and rotation, especially when weights on lower tiers are small or when ornaments ($p = 5, 7$) are suppressed. Our diagnostics (tier energies, phase polar plots) reduce but do not eliminate ambiguity. Hence point-estimates of operator order should be reported as gauge classes rather than unique words.

Discrete Operator Relaxations and Gradient Fidelity

Learning uses continuous relaxations for discrete choices (permutations π , topology moves $\hat{Q}$). Doubly-stochastic relaxations (e.g., Gumbel–Sinkhorn) and straight-through estimators introduce bias:

$$\mathbb{E}[\nabla_\pi \hat{R}] \neq \nabla_\pi \mathbb{E}[\hat{R}],$$

and annealing schedules may “freeze” into suboptimal permutations whose test-time projection degrades R . Similarly, surrogate losses for R_3 avoid arg discontinuities but can flatten gradients near phase lock. These issues limit convergence guarantees and complicate reproducibility when discrete search spaces are large.

Handling Incommensurate Periods at Scale

MOC assumes a finite lattice $\mathbb{T}_n$. Real data often mixes near-coprime or drifting periods, pushing n toward large lcm values. This increases memory and FFT cost, and it weakens CRT-tier separation when some prime powers are absent or very large. Approximations—windowed lattices, multi-rate tiling, rational approximants—help but introduce boundary artifacts and leakage that affect R_2 and R_3 . Heptadic overlays on scaffolds with $7 \nmid n$ are supported only as sparse gates, not true tiers, limiting formal symmetry.

Real-Time Constraints and Latency

Interactive uses (haptics, timers) require low latency. Rotation and spectral diagnostics favor FFT methods with $O(n \log n)$ cost and batch latency, conflicting with real-time feedback. Sliding updates for the DFT mitigate but still incur per-harmonic maintenance:

$$\widehat{X}_{t+1}[k] = \omega_n^{-k} \left(\widehat{X}_t[k] + x_{t+1} - x_{t+1-n} \right),$$

which is $O(n)$ per tick without subsampling. Phase-locked loop corrections improve alignment but add smoothing delay. In practice we must trade fidelity for responsiveness by (i) limiting active tiers, (ii) precompiling operator words, and (iii) scoring on downsampled combs. These choices can under-estimate syncopation effects and reduce sensitivity to weak tiers.

Additional Practical Limits

- **Energy/fairness constraints.** Hard caps on tier energies or fairness may render optimization infeasible; projection steps can distort noncommutative order effects.
- **Topology dynamics.** Relation operators $\hat{Q}$ that change incidence complicate caching and invalidate linear-time assumptions; current implementations restrict the frequency of topology moves.
- **Noise models.** The resonance functional presumes stationary noise within windows; heavy-tailed or bursty noise patterns bias R_1 and R_2 unless robust norms are used.

Overall, MOC is most reliable on moderate n with clear prime tiers, limited discrete search, and offline scoring. Streaming, highly incommensurate, or topology-heavy regimes expose the above limitations.

Future Work

Continuous-Time Extensions and Adelic Formulations

Continuous time. Replace the lattice $\mathbb{T}_n$ by the circle group $\mathbb{T} = \mathbb{R}/T\mathbb{Z}$ with signals in $L^2(\mathbb{T}, \mathbb{R}^k)$ or tempered distributions. Prime tiers become semigroups of flows

$$\mathcal{S}_p(\tau) = e^{\tau \mathcal{L}_p}, \quad \tau \geq 0,$$

where $\mathcal{L}_p$ generates p -rate refinement/rotation; discrete $\mathcal{S}_p, \mathcal{R}_{p^r}$ arise by sampling τ on rational grids. Comb masks generalize to Dirac combs $_{p^r}$; projectors Π_{p^r} become convolution with low-pass kernels supported on harmonics $\{k T/p^r\}$.

Adeles. Model multi-tier time as the restricted product

$$\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod_p' \mathbb{Q}_p,$$

with test space $\mathcal{S}(\mathbb{A}_{\mathbb{Q}})$ (Schwartz–Bruhat). Prime gates are local factors $\mathbf{1}_{\mathbb{Z}_p}$; the archimedean tier plays the role of continuous micro-timing. MOC operators become restricted tensor products $\widehat{\mathcal{P}} = \bigotimes_{p \leq \infty}' \widehat{\mathcal{P}}^{(p)}$. CRT on $\mathbb{Z}/n\mathbb{Z}$ lifts to adelic factorization; resonance extends via product integrals with separate weights for the real and p -adic parts.

Learning Relation Operators $\hat{Q}_p$ from Data

Objective. Jointly infer topology moves and state operators by maximizing R with regularizers on incidence complexity:

$$\max_{\hat{Q}, \Theta} R(\widehat{N}_{\Theta, \hat{Q}}[\mathcal{M}], D) - \lambda_{\text{topo}} \|\iota'\|_0.$$

Approaches.

- *Structural EM*: E-step proposes split/merge/fold candidates; M-step optimizes $\Theta = \{\alpha, \phi, \pi\}$.
- *Policy search*: reinforcement signal R guides a topology policy over {split, merge, fold, relabel} with acceptance tests on invariants.
- *Differentiable proxies*: soft-incidence tensors with entropic penalties; anneal to discrete ι .

Validation. Report topology edit distance, invariant preservation, and ablation ΔR per relation move.

Multi-Channel Vector Signals and Cross-Modal Coupling

Generalize $x : \mathbb{T}_n \rightarrow \mathbb{R}^k$ with coupling operators

$$\mathbf{C}_{p^r} : x \mapsto M_{p^r} * (\Delta_{p^r} x), \quad M_{p^r} \in \mathbb{R}^{k \times k},$$

to route tier events across channels (audio, haptics, visual). Enforce energy/fairness invariants per channel and cross-channel budgets (e.g., $\sum_c E_c \leq E_{\max}$, $F[\sigma] \geq F_{\min}$). Extend resonance to a weighted sum across modalities with modality weights ω_c and cross-coherence terms between channels.

Formal Category-Theoretic Semantics

Define a category **Mult**:

- *Objects*: multiplicity spaces $\mathcal{M} = (\mathcal{X}, H)$ with designated CRT tiers.

- *Morphisms*: gauge classes of operator words $[W] : \mathcal{M} \rightarrow \mathcal{M}'$ that preserve declared invariants.
- *Monoidal product*: $\otimes$ via CRT (coprime concatenation of tiers) yielding $\mathcal{M} \otimes \mathcal{M}'$.

Projectors Π_{p^r} are natural transformations; resonance R is a functor $\mathbf{Mult} \rightarrow [0, 1]$ stable under gauge. A 2-categorical layer can encode conjugation-by-gauge as 2-morphisms, clarifying identifiability and equivalence.

Tooling: Editor, Sequencer, Resonance Meter

Editor. Visual CRT canvas with tier toggles, operator graph view, gauge-fix controls, and invariant dashboards.

Sequencer. Prime-lane DAW: lanes for $p \in \{2, 3, 5, 7\}$, drag-drop $S, A, R, W, \hat{Q}$ blocks, real-time audition, export to MIDI/OSC/haptics.

Resonance meter. Live (R_1, R_2, R_3) with tier energy bars, phase polar plots, and ablation assistant that proposes minimal edits to raise R while respecting budgets.

Reproducibility. Project files store operator words, Θ , seeds, and gauge settings; one-click “freeze” produces deterministic renders and reports.

These trajectories extend MOC from discrete lattices to continuous and adelic settings, from fixed to learned relations, and from single-channel rhythms to cross-modal structures, while giving a categorical backbone and practical tools for design, analysis, and deployment.

Conclusion

Contributions and Findings

We presented the *Multiplicity Operator Calculus* (MOC), a prime-indexed, noncommutative framework acting jointly on signals and relations. The calculus supplies:

- operator families $(S_p, A_{p^r}, R_{p^r}, W_p; \hat{Q}_p)$ with explicit cross-prime noncommutation;
- CRT-tier projectors Π_{p^r} for analysis and control;
- a resonance functional $R = \lambda_1 R_1 + \lambda_2 R_2 + \lambda_3 R_3 \in [0, 1]$ that validates fit in time, spectrum, and phase;
- a worked $n = 108$ construction exposing the effect of operator order (ternary-first vs. binary-first);
- algorithms for learning continuous parameters and searching discrete structure, with reproducibility and invariants.

Experiments on metrical, ritual, and physiological data show consistent gains in harmonic lock and phase coherence over baselines, and ablations confirm the structural roles of $p = 3$ (phrasing) and $p = 2$ (micro-timing), with 5, 7 as stable ornaments.

Practical Guidance for Adoption

1. **Choose n by factorization.** Prefer n with meaningful prime powers; map data to $\mathbb{T}_n$ via robust period estimates.
2. **Start with a scaffold.** Build S_3^r then S_2^s for phrase-first timing; invert for syncopation.
3. **Gate and separate.** Set a monotone accent hierarchy on $\{p^r\}$; keep roles distinct using Π_{p^r} .
4. **Tune by R .** Optimize α and ϕ ; search π and word order with a small beam; always gauge-fix before reporting R .
5. **Respect budgets.** Enforce energy and fairness invariants; use ornaments (5, 7) at low gain and anti-coincide with spine tiers as needed.
6. **Reproduce.** Save seeds, configs, and the exact operator word; publish ablation deltas and tier diagnostics.

Open Problems and Next Steps

- **Theory.** Continuous-time and adelic extensions; category-theoretic semantics; links to modular forms and Hecke-style actions.
- **Learning.** Data-driven relation operators $\hat{Q}_p$; better relaxations for permutations; structured priors for operator order.
- **Identifiability.** Sharper invariants for distinguishing gauge-equivalent words; confidence sets over classes.
- **Scalability.** Multi-rate tilings for incommensurate periods; low-latency scoring for real-time feedback.
- **Multimodal.** Vector signals with cross-channel coupling; resonance across audio, haptics, and visual strata.
- **Tooling and benchmarks.** Open datasets with provenance, a prime-lane sequencer, and a standardized resonance meter.

MOC turns factorization into executable form: operators build structure, noncommutative order shapes feel, and resonance supplies evidence. The framework is compact, interpretable, and ready for controlled adoption in rhythm design, patterned scheduling, and pedagogy, with clear pathways to deeper mathematics and broader applications.

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Appendix A. Full Derivations for Projector Identities

A.1 Definition and Group-Average Form

Fix $n \in \mathbb{N}$ and $x \in \mathcal{X}_n = \{x : \mathbb{T}_n \rightarrow \mathbb{R}^k\}$. For a prime power $p^r \mid n$ define

$$(\Pi_{p^r} x)(t) = \frac{1}{p^r} \sum_{u=0}^{p^r-1} x\left(t + u \frac{n}{p^r}\right), \quad t \in \mathbb{T}_n.$$

Let $\mathbf{R}_{p^r}$ denote rotation by $\tau = \frac{n}{p^r}$ ticks, i.e. $(\mathbf{R}_{p^r} x)(t) = x(t + \tau)$. Then

$$\Pi_{p^r} = \frac{1}{p^r} \sum_{u=0}^{p^r-1} \mathbf{R}_{p^r}^u$$

is the Reynolds (orbit-averaging) operator for the cyclic subgroup $G_{p^r} = \langle \mathbf{R}_{p^r} \rangle$ of order p^r .

A.2 Frequency-Domain Characterization

Let $\widehat{x}[k] = \sum_{t=0}^{n-1} x(t) \omega_n^{-kt}$ be the n -point DFT, $\omega_n = e^{2\pi i/n}$. Then

$$\widehat{\Pi_{p^r} x}[k] = \frac{1}{p^r} \sum_{u=0}^{p^r-1} \omega_n^{-kun/p^r} \widehat{x}[k] = \left(\frac{1}{p^r} \sum_{u=0}^{p^r-1} e^{-2\pi i k u / p^r} \right) \widehat{x}[k].$$

Hence

$$\widehat{\Pi_{p^r} x}[k] = \mathbf{1}_{p^r|k} \widehat{x}[k],$$

i.e. Π_{p^r} is the orthogonal projector onto the subspace spanned by harmonics with indices k that are multiples of p^r . Equivalently, $\text{range}(\Pi_{p^r}) = \{x : \mathbf{R}_{p^r} x = x\}$.

A.3 Idempotence, Self-Adjointness, and Orthogonality

Idempotence. By group averaging or masks:

$$\Pi_{p^r}^2 = \left(\frac{1}{p^r} \sum_u \mathbf{R}_{p^r}^u \right) \left(\frac{1}{p^r} \sum_v \mathbf{R}_{p^r}^v \right) = \frac{1}{p^r} \sum_w \mathbf{R}_{p^r}^w = \Pi_{p^r},$$

$$\text{or } \widehat{\Pi^2}[k] = \mathbf{1}_{p^r|k}^2 \widehat{x}[k] = \mathbf{1}_{p^r|k} \widehat{x}[k].$$

Self-adjointness. With the standard inner product $\langle x, y \rangle = \sum_t x(t)^\top y(t)$, rotations are unitary, so Π_{p^r} is self-adjoint as an average of unitaries. In frequency, it is a real diagonal mask.

Orthogonal decomposition (same prime). For $a \leq b$,

$$\Pi_{p^a} \Pi_{p^b} = \Pi_{p^b}, \quad \Pi_{p^a} - \Pi_{p^b} \text{ projects onto } \{k : p^a \mid k, p^b \nmid k\}.$$

Mixed primes. For coprime $p \neq q$,

$$\Pi_{p^r} \Pi_{q^s} = \Pi_{q^s} \Pi_{p^r} = \Pi_{p^r q^s},$$

$$\text{since } \mathbf{1}_{p^r|k} \mathbf{1}_{q^s|k} = \mathbf{1}_{p^r q^s|k}.$$

A.4 Commutation with Rotations

For any $\phi \in \mathbb{Z}$,

$$\Pi_{p^r} \mathbf{R}_{p^r}^\phi = \frac{1}{p^r} \sum_u \mathbf{R}_{p^r}^{u+\phi} = \Pi_{p^r}, \quad \mathbf{R}_{p^r}^\phi \Pi_{p^r} = \Pi_{p^r},$$

so $\text{range}(\Pi_{p^r})$ is fixed-pointwise invariant under $\mathbf{R}_{p^r}$.

More generally, rotations commute as operators; hence Π_{p^r} commutes with any $\mathbf{R}_m$:

$$\mathbf{R}_m \Pi_{p^r} = \Pi_{p^r} \mathbf{R}_m,$$

but only for $m = p^r$ do we have $\Pi_{p^r} \mathbf{R}_m = \Pi_{p^r}$.

A.5 Interaction with Accent Gates and Spike Projectors

Define the spike (multiplicative) gate Δ_d by $(\Delta_d x)(t) = [d \mid t] x(t)$ and the additive accent $A_d^\alpha : x \mapsto x + \alpha [d \mid t] e_1$.

Fourier of the comb.

$$\widehat{[d \mid \cdot]}[k] = \sum_{m=0}^{n/d-1} \omega_n^{-kmd} = \frac{n}{d} \mathbf{1}_{n/d \mid k}.$$

Noncommutation in general.

$$[\Pi_{p^r}, A_{p^s}^\alpha]x = \alpha \left(\Pi_{p^r} [p^s \mid \cdot] - [p^s \mid \cdot] \right) e_1,$$

which vanishes iff $\Pi_{p^r} [p^s \mid \cdot] = [p^s \mid \cdot]$.

Commutation condition. $\Pi_{p^r} [p^s \mid \cdot] = [p^s \mid \cdot]$ holds iff the comb $[p^s \mid t]$ is invariant under R_{p^r} , i.e.

$$\frac{n}{p^r} \equiv 0 \pmod{p^s} \iff p^{r+s} \mid n.$$

Equivalently, if $v_p(n) \geq r + s$. In that case,

$$\Pi_{p^r} A_{p^s}^\alpha = A_{p^s}^\alpha \Pi_{p^r}.$$

Otherwise, they do not commute.

Spike projector composition. In frequency, $\widehat{\Delta_{p^s} x} = \widehat{x} * \widehat{[p^s \mid \cdot]}$, so

$$\widehat{\Pi_{p^r} \Delta_{p^s} x}[k] = \mathbf{1}_{p^r \mid k} (\widehat{x} * \widehat{[p^s \mid \cdot]})(k),$$

which differs from $\widehat{\Delta_{p^s} \Pi_{p^r} x}[k]$ unless the support of $\widehat{[p^s \mid \cdot]}$ lies within $\{k : p^r \mid k\}$, i.e. unless n/p^s is a multiple of p^r .

A.6 Subdivision and Projector Interchange

Let $S_q : \mathcal{X}_n \rightarrow \mathcal{X}_{qn}$ be $(S_q x)(qt + u) = x(t)$. Consider Π_{p^r} on the refined lattice $\mathbb{T}_{qn}$. In frequency,

$$\widehat{S_q x}[k] = D_q \left(\frac{2\pi k}{qn} \right) \widehat{x}[k \bmod n],$$

where D_q is the Dirichlet kernel. Hence

$$\widehat{\Pi_{p^r}^{(qn)} S_q x}[k] = \mathbf{1}_{p^r \mid k} D_q \left(\frac{2\pi k}{qn} \right) \widehat{x}[k \bmod n].$$

On the coarse lattice, define the induced projector

$$(\Pi_{p^r}^{\downarrow q} x)^\wedge[\ell] = \mathbf{1}_{p^r | (q\ell)} \widehat{x}[\ell] = \mathbf{1}_{p^{r-v_p(q)} | \ell} \widehat{x}[\ell] \quad (\text{by } v_p \text{ arithmetic, with the convention } p^{-a} \nmid \ell \text{ if } a > 0).$$

Then

$$\widehat{\mathbf{S}_q \Pi_{p^r}^{\downarrow q} x}[k] = D_q \left(\frac{2\pi k}{qn} \right) \mathbf{1}_{p^{r-v_p(q)} | (k \bmod n)} \widehat{x}[k \bmod n].$$

Comparing the two shows

$$\Pi_{p^r}^{(qn)} \mathbf{S}_q = \mathbf{S}_q \Pi_{p^{\max(0, r-v_p(q))}}^{\downarrow q}.$$

Special cases:

- If $p \nmid q$, then $v_p(q) = 0$ and $\Pi_{p^r}^{(qn)} \mathbf{S}_q = \mathbf{S}_q \Pi_{p^r}$.
- If $p^r \mid q$, then $\Pi_{p^r}^{(qn)} \mathbf{S}_q = \mathbf{S}_q$ (the refined lattice is already invariant at level p^r).

A.7 Permutations Inside Cells

Let $\mathbf{W}_q^\pi$ permute the q children within each parent cell: $(\mathbf{W}_q^\pi x)(qt + u) = x(qt + \pi(u))$. In frequency, $\mathbf{W}_q^\pi$ mixes the q -tuple of coefficients in each residue class $k \equiv r \pmod{q}$ by a unitary $U_\pi(r)$ that depends on r and π . Hence

$$\widehat{\Pi_{p^r} \mathbf{W}_q^\pi x}[k] = \mathbf{1}_{p^r | k} \sum_{h \equiv k \pmod{q}} [U_\pi(k \bmod q)]_{k,h} \widehat{x}[h],$$

while

$$\widehat{\mathbf{W}_q^\pi \Pi_{p^r} x}[k] = \sum_{h \equiv k \pmod{q}} [U_\pi(k \bmod q)]_{k,h} \mathbf{1}_{p^r | h} \widehat{x}[h].$$

Unless p^r divides all h in the residue class $k \bmod q$ (rare), the two differ; thus $[\Pi_{p^r}, \mathbf{W}_q^\pi] \neq 0$ in general.

A.8 Equivalent Characterizations

For $p^r \mid n$, the following are equivalent for $y \in \mathcal{X}_n$:

1. $\Pi_{p^r} y = y$.
2. $\mathbf{R}_{p^r} y = y$.
3. $\widehat{y}[k] = 0$ unless $p^r \mid k$.
4. y factors through the CRT projection $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/(n/p^r)\mathbb{Z}$, i.e. $y(t)$ depends only on $t \pmod{n/p^r}$.

Proof. (1) $\Leftrightarrow$ (2): Π is the Reynolds projector onto fixed points. (2) $\Rightarrow$ (3): rotation invariance imposes $e^{-2\pi i k/p^r} = 1$. (3) $\Rightarrow$ (4): only harmonics compatible with period n/p^r remain. (4) $\Rightarrow$ (2): periodicity implies invariance under shift by n/p^r .

A.9 Summary of Identities

For $p^r \mid n$, primes $p \neq q$, and any x :

$$\begin{aligned} \Pi_{p^r}^2 &= \Pi_{p^r}, & \Pi_{p^r}^\top &= \Pi_{p^r}, & \widehat{\Pi_{p^r} x}[k] &= \mathbf{1}_{p^r \mid k} \widehat{x}[k], \\ \Pi_{p^a} \Pi_{p^b} &= \Pi_{p^{\max(a,b)}}, & \Pi_{p^r} \Pi_{q^s} &= \Pi_{q^s} \Pi_{p^r} = \Pi_{p^r q^s}, \\ \Pi_{p^r} \mathbf{R}_{p^r}^\phi &= \mathbf{R}_{p^r}^\phi \Pi_{p^r} = \Pi_{p^r}, & [\Pi_{p^r}, \mathbf{R}_m] &= 0 \quad \forall m, \\ \Pi_{p^r} \mathbf{A}_{p^s}^\alpha &= \mathbf{A}_{p^s}^\alpha \Pi_{p^r} \quad \text{iff } v_p(n) \geq r+s, \quad \text{else } [\Pi_{p^r}, \mathbf{A}_{p^s}^\alpha] \neq 0, \\ \Pi_{p^r}^{(qn)} \mathbf{S}_q &= \mathbf{S}_q \Pi_{p^{\max(0, r-v_p(q))}}^{\downarrow q}, & [\Pi_{p^r}, \mathbf{W}_q^\pi] &\neq 0 \text{ in general.} \end{aligned}$$

These equalities formalize the intuitive picture: Π_{p^r} keeps precisely the content invariant under rotation by n/p^r , composes monotonically across tiers, commutes with rotations and with accents only under a clear p -adic condition, intertwines predictably with subdivision, and generally conflicts with within-cell permutations that remix residue classes.

Appendix A. Full Derivations for Projector Identities

A.1 Definition and Group-Average Form

Fix $n \in \mathbb{N}$ and $x \in \mathcal{X}_n = \{x : \mathbb{T}_n \rightarrow \mathbb{R}^k\}$. For a prime power $p^r \mid n$ define

$$(\Pi_{p^r} x)(t) = \frac{1}{p^r} \sum_{u=0}^{p^r-1} x\left(t + u \frac{n}{p^r}\right), \quad t \in \mathbb{T}_n.$$

Let $\mathbf{R}_{p^r}$ denote rotation by $\tau = \frac{n}{p^r}$ ticks, i.e. $(\mathbf{R}_{p^r} x)(t) = x(t + \tau)$. Then

$$\Pi_{p^r} = \frac{1}{p^r} \sum_{u=0}^{p^r-1} \mathbf{R}_{p^r}^u$$

is the Reynolds (orbit-averaging) operator for the cyclic subgroup $G_{p^r} = \langle \mathbf{R}_{p^r} \rangle$ of order p^r .

A.2 Frequency-Domain Characterization

Let $\widehat{x}[k] = \sum_{t=0}^{n-1} x(t) \omega_n^{-kt}$ be the n -point DFT, $\omega_n = e^{2\pi i/n}$. Then

$$\widehat{\Pi_{p^r} x}[k] = \frac{1}{p^r} \sum_{u=0}^{p^r-1} \omega_n^{-kun/p^r} \widehat{x}[k] = \left(\frac{1}{p^r} \sum_{u=0}^{p^r-1} e^{-2\pi i k u / p^r} \right) \widehat{x}[k].$$

Hence

$$\widehat{\Pi_{p^r} x}[k] = \mathbf{1}_{p^r \mid k} \widehat{x}[k],$$

i.e. Π_{p^r} is the orthogonal projector onto the subspace spanned by harmonics with indices k that are multiples of p^r . Equivalently, $\text{range}(\Pi_{p^r}) = \{x : \mathbf{R}_{p^r} x = x\}$.

A.3 Idempotence, Self-Adjointness, and Orthogonality

Idempotence. By group averaging or masks:

$$\Pi_{p^r}^2 = \left(\frac{1}{p^r} \sum_u \mathbf{R}_{p^r}^u \right) \left(\frac{1}{p^r} \sum_v \mathbf{R}_{p^r}^v \right) = \frac{1}{p^r} \sum_w \mathbf{R}_{p^r}^w = \Pi_{p^r},$$

$$\text{or } \widehat{\Pi^2}[k] = \mathbf{1}_{p^r|k}^2 \widehat{x}[k] = \mathbf{1}_{p^r|k} \widehat{x}[k].$$

Self-adjointness. With the standard inner product $\langle x, y \rangle = \sum_t x(t)^\top y(t)$, rotations are unitary, so Π_{p^r} is self-adjoint as an average of unitaries. In frequency, it is a real diagonal mask.

Orthogonal decomposition (same prime). For $a \leq b$,

$$\Pi_{p^a} \Pi_{p^b} = \Pi_{p^b}, \quad \Pi_{p^a} - \Pi_{p^b} \text{ projects onto } \{k : p^a \mid k, p^b \nmid k\}.$$

Mixed primes. For coprime $p \neq q$,

$$\Pi_{p^r} \Pi_{q^s} = \Pi_{q^s} \Pi_{p^r} = \Pi_{p^r q^s},$$

$$\text{since } \mathbf{1}_{p^r|k} \mathbf{1}_{q^s|k} = \mathbf{1}_{p^r q^s|k}.$$

A.4 Commutation with Rotations

For any $\phi \in \mathbb{Z}$,

$$\Pi_{p^r} \mathbf{R}_{p^r}^\phi = \frac{1}{p^r} \sum_u \mathbf{R}_{p^r}^{u+\phi} = \Pi_{p^r}, \quad \mathbf{R}_{p^r}^\phi \Pi_{p^r} = \Pi_{p^r},$$

so $\text{range}(\Pi_{p^r})$ is fixed-pointwise invariant under $\mathbf{R}_{p^r}$.

More generally, rotations commute as operators; hence Π_{p^r} commutes with any $\mathbf{R}_m$:

$$\mathbf{R}_m \Pi_{p^r} = \Pi_{p^r} \mathbf{R}_m,$$

but only for $m = p^r$ do we have $\Pi_{p^r} \mathbf{R}_m = \Pi_{p^r}$.

A.5 Interaction with Accent Gates and Spike Projectors

Define the spike (multiplicative) gate Δ_d by $(\Delta_d x)(t) = [d \mid t] x(t)$ and the additive accent $\mathbf{A}_d^\alpha : x \mapsto x + \alpha [d \mid t] e_1$.

Fourier of the comb.

$$\widehat{[d \mid \cdot]}[k] = \sum_{m=0}^{n/d-1} \omega_n^{-kmd} = \frac{n}{d} \mathbf{1}_{n/d \mid k}.$$

Noncommutation in general.

$$[\Pi_{p^r}, \mathbf{A}_{p^s}^\alpha]x = \alpha \left(\Pi_{p^r} [p^s \mid \cdot] - [p^s \mid \cdot] \right) e_1,$$

which vanishes iff $\Pi_{p^r} [p^s \mid \cdot] = [p^s \mid \cdot]$.

Commutation condition. $\Pi_{p^r} [p^s \mid \cdot] = [p^s \mid \cdot]$ holds iff the comb $[p^s \mid t]$ is invariant under $\mathbf{R}_{p^r}$, i.e.

$$\frac{n}{p^r} \equiv 0 \pmod{p^s} \iff p^{r+s} \mid n.$$

Equivalently, if $v_p(n) \geq r + s$. In that case,

$$\Pi_{p^r} \mathbf{A}_{p^s}^\alpha = \mathbf{A}_{p^s}^\alpha \Pi_{p^r}.$$

Otherwise, they do not commute.

Spike projector composition. In frequency, $\widehat{\Delta_{p^s} x} = \widehat{x} * \widehat{[p^s \mid \cdot]}$, so

$$\widehat{\Pi_{p^r} \Delta_{p^s} x}[k] = \mathbf{1}_{p^r \mid k} (\widehat{x} * \widehat{[p^s \mid \cdot]})(k),$$

which differs from $\widehat{\Delta_{p^s} \Pi_{p^r} x}[k]$ unless the support of $\widehat{[p^s \mid \cdot]}$ lies within $\{k : p^r \mid k\}$, i.e. unless n/p^s is a multiple of p^r .

A.6 Subdivision and Projector Interchange

Let $\mathbf{S}_q : \mathcal{X}_n \rightarrow \mathcal{X}_{qn}$ be $(\mathbf{S}_q x)(qt + u) = x(t)$. Consider Π_{p^r} on the refined lattice $\mathbb{T}_{qn}$. In frequency,

$$\widehat{\mathbf{S}_q x}[k] = D_q \left(\frac{2\pi k}{qn} \right) \widehat{x}[k \bmod n],$$

where D_q is the Dirichlet kernel. Hence

$$\widehat{\Pi_{p^r}^{(qn)} \mathbf{S}_q x}[k] = \mathbf{1}_{p^r \mid k} D_q \left(\frac{2\pi k}{qn} \right) \widehat{x}[k \bmod n].$$

On the coarse lattice, define the induced projector

$$(\Pi_{p^r}^{\downarrow q} x)^\sim[\ell] = \mathbf{1}_{p^r \mid (q\ell)} \widehat{x}[\ell] = \mathbf{1}_{p^{r-v_p(q)} \mid \ell} \widehat{x}[\ell] \quad (\text{by } v_p \text{ arithmetic, with the convention } p^{-a} \nmid \ell \text{ if } a > 0).$$

Then

$$\widehat{\mathbf{S}_q \Pi_{p^r}^{\downarrow q} x}[k] = D_q \left(\frac{2\pi k}{qn} \right) \mathbf{1}_{p^{r-v_p(q)} | (k \bmod n)} \widehat{x}[k \bmod n].$$

Comparing the two shows

$$\Pi_{p^r}^{(qn)} \mathbf{S}_q = \mathbf{S}_q \Pi_{p^{\max(0, r-v_p(q))}}^{\downarrow q}.$$

Special cases:

- If $p \nmid q$, then $v_p(q) = 0$ and $\Pi_{p^r}^{(qn)} \mathbf{S}_q = \mathbf{S}_q \Pi_{p^r}$.
- If $p^r \mid q$, then $\Pi_{p^r}^{(qn)} \mathbf{S}_q = \mathbf{S}_q$ (the refined lattice is already invariant at level p^r).

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Let $\mathbf{W}_q^\pi$ permute the q children within each parent cell: $(\mathbf{W}_q^\pi x)(qt + u) = x(qt + \pi(u))$. In frequency, $\mathbf{W}_q^\pi$ mixes the q -tuple of coefficients in each residue class $k \equiv r \pmod{q}$ by a unitary $U_\pi(r)$ that depends on r and π . Hence

$$\widehat{\Pi_{p^r} \mathbf{W}_q^\pi x}[k] = \mathbf{1}_{p^r | k} \sum_{h \equiv k \pmod{q}} [U_\pi(k \bmod q)]_{k,h} \widehat{x}[h],$$

while

$$\widehat{\mathbf{W}_q^\pi \Pi_{p^r} x}[k] = \sum_{h \equiv k \pmod{q}} [U_\pi(k \bmod q)]_{k,h} \mathbf{1}_{p^r | h} \widehat{x}[h].$$

Unless p^r divides all h in the residue class $k \bmod q$ (rare), the two differ; thus $[\Pi_{p^r}, \mathbf{W}_q^\pi] \neq 0$ in general.

A.8 Equivalent Characterizations

For $p^r \mid n$, the following are equivalent for $y \in \mathcal{X}_n$:

1. $\Pi_{p^r} y = y$.
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Proof. (1) $\Leftrightarrow$ (2): Π is the Reynolds projector onto fixed points. (2) $\Rightarrow$ (3): rotation invariance imposes $e^{-2\pi i k/p^r} = 1$. (3) $\Rightarrow$ (4): only harmonics compatible with period n/p^r remain. (4) $\Rightarrow$ (2): periodicity implies invariance under shift by n/p^r .

A.9 Summary of Identities

For $p^r \mid n$, primes $p \neq q$, and any x :

$$\Pi_{p^r}^2 = \Pi_{p^r}, \quad \Pi_{p^r}^\top = \Pi_{p^r}, \quad \widehat{\Pi_{p^r} x}[k] = \mathbf{1}_{p^r | k} \widehat{x}[k],$$

$$\begin{aligned}
\Pi_{p^a} \Pi_{p^b} &= \Pi_{p^{\max(a,b)}}, & \Pi_{p^r} \Pi_{q^s} &= \Pi_{q^s} \Pi_{p^r} = \Pi_{p^r q^s}, \\
\Pi_{p^r} \mathbf{R}_{p^r}^\phi &= \mathbf{R}_{p^r}^\phi \Pi_{p^r} = \Pi_{p^r}, & [\Pi_{p^r}, \mathbf{R}_m] &= 0 \quad \forall m, \\
\Pi_{p^r} \mathbf{A}_{p^s}^\alpha &= \mathbf{A}_{p^s}^\alpha \Pi_{p^r} \quad \text{iff } v_p(n) \geq r+s, \quad \text{else } [\Pi_{p^r}, \mathbf{A}_{p^s}^\alpha] \neq 0, \\
\Pi_{p^r}^{(qn)} \mathbf{S}_q &= \mathbf{S}_q \Pi_{p^{\max(0, r-v_p(q))}}^{\downarrow q}, & [\Pi_{p^r}, \mathbf{W}_q^\pi] &\neq 0 \text{ in general.}
\end{aligned}$$

These equalities formalize the intuitive picture: Π_{p^r} keeps precisely the content invariant under rotation by n/p^r , composes monotonically across tiers, commutes with rotations and with accents only under a clear p -adic condition, intertwines predictably with subdivision, and generally conflicts with within-cell permutations that remix residue classes.

Appendix B. Additional Operator Identities and Counterexamples to Commutation

B.1 Algebraic Identities (State Layer)

Let $n \in \mathbb{N}$, p, q primes, and $r, s \geq 1$ with $p^r, q^s \mid n$. Operators act on $\mathcal{X}_n$ unless noted.

Rotations. For any integers ϕ, ψ ,

$$\mathbf{R}_{p^r}^\phi \mathbf{R}_{p^r}^\psi = \mathbf{R}_{p^r}^{\phi+\psi}, \quad \mathbf{R}_{p^r}^\phi \mathbf{R}_{q^s}^\psi = \mathbf{R}_{q^s}^\psi \mathbf{R}_{p^r}^\phi \quad (= \mathbf{R}_n^{\phi n/p^r + \psi n/q^s}).$$

Additive accents. Using $D_d(t) = [d \mid t]$ and channel e_1 ,

$$\mathbf{A}_d^\alpha x = x + \alpha D_d e_1^\top, \quad \mathbf{A}_{d_1}^{\alpha_1} \mathbf{A}_{d_2}^{\alpha_2} = \mathbf{A}_{d_2}^{\alpha_2} \mathbf{A}_{d_1}^{\alpha_1}, \quad \mathbf{A}_d^\alpha \mathbf{A}_d^\beta = \mathbf{A}_d^{\alpha+\beta}.$$

Spike gates. $\Delta_d x = D_d \odot x$ (Hadamard product) obeys

$$\Delta_a \Delta_b = \Delta_{\text{lcm}(a,b)}, \quad \Delta_d \mathbf{R}_m = \mathbf{R}_m \Delta_d \quad \text{iff } m \equiv 0 \pmod{d}.$$

Subdivision and rotations (lifted identity). Let $\mathbf{S}_p : \mathcal{X}_n \rightarrow \mathcal{X}_{pn}$, and let $\mathbf{R}_{p^r}^{\phi, (pn)}$ denote rotation on the refined lattice. Then

$$\mathbf{R}_{p^r}^{\phi, (pn)} \mathbf{S}_p = \mathbf{S}_p \mathbf{R}_{p^r}^\phi \quad (p^r \mid n).$$

Proof: both sides shift by $\tau = \phi n/p^r$ at the parent resolution, replicated to children.

Subdivision and accents (lift rule).

$$\mathbf{S}_p \mathbf{A}_{q^s}^\alpha = (\mathbf{A}_{q^s}^\alpha)^{\uparrow p} \mathbf{S}_p, \quad \text{where } (\mathbf{A}_{q^s}^\alpha)^{\uparrow p} \text{ copies the comb into each of the } p \text{ subticks.}$$

Within-cell permutations. For $\pi_1, \pi_2 \in S_p$,

$$W_p^{\pi_1} W_p^{\pi_2} = W_p^{\pi_1 \circ \pi_2}, \quad (W_p^\pi)^{-1} = W_p^{\pi^{-1}}.$$

Moreover,

$$R_{n/p}^m W_p^\pi = W_p^\pi R_{n/p}^m \quad \text{for all } m \in \mathbb{Z},$$

since $R_{n/p}$ shifts by one parent cell; but for general τ not divisible by n/p commutation fails (see B.3.2).

Phase-offset accents. Define $A_d^{\alpha, \delta} : x \mapsto x + \alpha \mathbf{1}_{t \equiv \delta \pmod{d}} e_1^\top$. Then

$$R_d^\phi A_d^{\alpha, 0} = A_d^{\alpha, \phi} R_d^\phi.$$

This reparameterization isolates noncommutation as a phase change of the gate.

B.2 Identities Involving Projectors (Summary)

For completeness (see Appendix A for proofs):

$$\Pi_{p^r}^2 = \Pi_{p^r}, \quad \Pi_{p^r} \Pi_{q^s} = \Pi_{p^r q^s} \quad (p \neq q), \quad \Pi_{p^a} \Pi_{p^b} = \Pi_{p^{\max(a,b)}},$$

$$\Pi_{p^r} R_m = R_m \Pi_{p^r}, \quad \Pi_{p^r} A_{p^s}^\alpha = A_{p^s}^\alpha \Pi_{p^r} \text{ iff } v_p(n) \geq r + s,$$

and generally $[\Pi_{p^r}, W_q^\pi] \neq 0$.

B.3 Counterexamples to Commutation (Concrete)

We use $n = 12$ or $n = 18$ for explicit vectors; take a single channel and denote the Kronecker delta by δ_t .

B.3.1 $[W_2^{(01)}, A_3^\alpha] \neq 0$ on $n = 12$. Let $x \equiv 0$, A_3^α place ones at $t \in \{0, 3, 6, 9\}$. With $W_2^{(01)}$ swapping within pairs $(0, 1), (2, 3), \dots$,

$$W_2^{(01)} A_3^\alpha x = \sum_{t \in \{1, 2, 7, 8\}} \delta_t, \quad A_3^\alpha W_2^{(01)} x = \sum_{t \in \{0, 3, 6, 9\}} \delta_t.$$

The supports differ; hence noncommutation.

B.3.2 $[R_3^1, W_3^{(012)}] \neq 0$ on $n = 12$. Cells of size 3 are $(0..2), (3..5), (6..8), (9..11)$. Let $x = \delta_0$.

$$W_3^{(012)} x = \delta_1, \quad R_3^1(\delta_1) = \delta_{1+4} = \delta_5.$$

Conversely,

$$R_3^1 x = \delta_4, \quad W_3^{(012)}(\delta_4) = \delta_3.$$

Outputs differ ($t = 5$ vs $t = 3$).

B.3.3 $[S_2, A_3^\alpha] \neq 0$ **without lift.** $n = 12, x \equiv 0$. Then

$$S_2 A_3^\alpha x : \text{ones at } t \in \{0, 6, 12, 18\} \pmod{24},$$

but

$$A_3^\alpha S_2 x : \text{ones at } t \in \{0, 3, 6, 9, 12, 15, 18, 21\}.$$

They differ unless the lift rule $(\cdot)^{\uparrow 2}$ is used.

B.3.4 $[\Pi_3, W_2^{(01)}] \neq 0$ **on** $n = 12$. Let $x = \delta_0$. Then

$$\Pi_3 W_2^{(01)} x = \Pi_3(\delta_1) = \frac{1}{3}(\delta_1 + \delta_5 + \delta_9),$$

while

$$W_2^{(01)} \Pi_3 x = W_2^{(01)} \left(\frac{1}{3}(\delta_0 + \delta_4 + \delta_8) \right) = \frac{1}{3}(\delta_1 + \delta_5 + \delta_9).$$

Here they coincide for δ_0 , but choose $x = \delta_2$:

$$\Pi_3 W_2^{(01)} \delta_2 = \Pi_3(\delta_3) = \frac{1}{3}(\delta_3 + \delta_7 + \delta_{11}),$$

$$W_2^{(01)} \Pi_3 \delta_2 = W_2^{(01)} \left(\frac{1}{3}(\delta_2 + \delta_6 + \delta_{10}) \right) = \frac{1}{3}(\delta_3 + \delta_7 + \delta_{11}).$$

These also match; noncommutation appears for general mixtures. Take $x = \delta_1 + \delta_2$:

$$\Pi_3 W_2^{(01)} x = \frac{1}{3}(\delta_0 + \delta_4 + \delta_8) + \frac{1}{3}(\delta_3 + \delta_7 + \delta_{11}),$$

$$W_2^{(01)} \Pi_3 x = \frac{1}{3}(\delta_1 + \delta_5 + \delta_9) + \frac{1}{3}(\delta_3 + \delta_7 + \delta_{11}),$$

which differ.

B.3.5 $[R_{p^r}^\phi, A_{p^r}^\alpha] \neq 0$ **unless phase-offset is used.** On $n = 108, p^r = 27, \phi = 1$:

$$R_{27}^1 A_{27}^\alpha x = R_{27}^1 x + \alpha \sum_{t \in \{0, 27, 54, 81\}} \delta_{t+4},$$

$$A_{27}^\alpha R_{27}^1 x = R_{27}^1 x + \alpha \sum_{t \in \{0, 27, 54, 81\}} \delta_t.$$

Not equal; but $R_{27}^1 A_{27}^{\alpha, 0} = A_{27}^{\alpha, 1} R_{27}^1$ holds.

B.4 Identities Involving Relation Operators

Let $H = (V, E, \iota)$, and $\hat{Q}_p^{\text{split}}$ duplicate each edge e into $e^{(0)}, \dots, e^{(p-1)}$ with copied incidence.

Split vs. additive accents on edge-aggregated channels. Suppose an edge channel aggregates vertex signals by sum, and A_d^α adds weight on a vertex channel at gate times. Then

$$\hat{Q}_p^{\text{split}} \circ A_d^\alpha \neq A_d^\alpha \circ \hat{Q}_p^{\text{split}}$$

in general, because splitting before accent duplicates the aggregation targets and may double-count at gate ticks; splitting after accent preserves a single aggregation per original edge.

Example (noncommutation). Let $V = \{v_1, v_2\}$, $E = \{e\}$ with $\iota(e) = \{v_1, v_2\}$, $n = 6$, and let the edge signal be $\tau(e, t) = \sigma(v_1, t) + \sigma(v_2, t)$. Start with $\sigma \equiv 0$, apply A_3^α to $\sigma(v_1, \cdot)$.

$$(\hat{Q}_2^{\text{split}} \circ A_3^\alpha) : \tau(e^{(0)}, t) = \tau(e^{(1)}, t) = \alpha [3 \mid t],$$

total edge mass 2α per gate.

$$(A_3^\alpha \circ \hat{Q}_2^{\text{split}}) : \text{accent on } v_1 \text{ once, then sum across split edges} \Rightarrow \text{total } \alpha \text{ per gate.}$$

Thus $[\hat{Q}_2^{\text{split}}, A_3^\alpha] \neq 0$.

Fold vs. rotation. If $\hat{Q}_p^{\text{fold}}$ quotients p cells by cyclic symmetry, then

$$\hat{Q}_p^{\text{fold}} R_{n/p}^1 = \hat{Q}_p^{\text{fold}}, \quad R_{n/p}^1 \hat{Q}_p^{\text{fold}} \neq \hat{Q}_p^{\text{fold}}$$

on the pre-folded space, since the quotient kills the degree of freedom that the rotation would otherwise act on.

B.5 Mixed-Prime Wreath Interactions

Assume n divisible by pq . Define W_p and W_q using contiguous blockings of size p and q respectively. Then, in general,

$$[W_p^\pi, W_q^\rho] \neq 0.$$

Counterexample ($n = 6$, $p = 2$, $q = 3$). Let $x = (x_0, \dots, x_5)$, $\pi = (0 \ 1)$, $\rho = (0 \ 1 \ 2)$. Compute within blocks:

$$W_2^\pi W_3^\rho x = \text{swap within } (0, 1), (2, 3), (4, 5) \text{ after cycling } (0, 1, 2), (3, 4, 5),$$

$$W_3^\rho W_2^\pi x = \text{cycle within } (0, 1, 2), (3, 4, 5) \text{ after swapping the pairs.}$$

Coordinate expressions differ unless x satisfies special symmetries.

B.6 Round-Up: When Do Things Commute?

Pair	Condition	Outcome
R, R	always	commute
A_{d_1}, A_{d_2}	always (additive)	commute
Δ_a, Δ_b	always	$\Delta_{\text{lcm}(a,b)}$
$R_{n/p}, W_p$	shift by parent cell	commute
R_{p^r}, A_{p^r}	with phase-offset accent	$RA^{(\delta)} = A^{(\delta+\phi)}R$
Π_{p^r}, A_{p^s}	$v_p(n) \geq r + s$	commute
Π_{p^r}, W_q	generic	do <i>not</i> commute
S_p, R_{p^r}	$p^r \mid n$	commute after lift
S_p, A_{q^s}	generic	lift rule $(\cdot)^{\uparrow p}$ required
$\hat{Q}_p^{\text{split}}, A_d$	generic	do <i>not</i> commute

B.7 Practical Note

In implementations, encode identities as rewrite rules in the execution planner:

$$R_{p^r}^\phi \circ A_{p^r}^{\alpha,0} \Rightarrow A_{p^r}^{\alpha,\phi} \circ R_{p^r}^\phi, \quad S_p \circ A_{q^s}^\alpha \Rightarrow (A_{q^s}^\alpha)^{\uparrow p} \circ S_p,$$

$$W_p^{\pi_1} \circ W_p^{\pi_2} \Rightarrow W_p^{\pi_1 \circ \pi_2}, \quad \Pi_{p^r} \circ \Pi_{q^s} \Rightarrow \Pi_{p^r q^s} \ (p \neq q).$$

Counterexample tests (small n harnesses) should be part of unit suites to guard against accidental operator reordering during kernel fusion.

Appendix C. Extended Pseudocode and APIs

C.1 Core Types and Layout

Scalars and aliases

```

Tick    := int           # 0..n-1
Index   := int
Real    := float
Vec[T]  := array of T
Mat[T]  := 2D array of T
Bool    := {True, False}

```

Lattice and signals

```

Lattice := struct { n: int }           # length

```

```

Signal := struct { X: Mat[Real], n: int, k: int } # shape (n, k), row-major

# Hypergraph
Hyper := struct {
  V: int, E: int,
  edge_ptr: Vec[int],      # CSR: size E+1
  edge_vtx: Vec[int],      # concatenated vertex ids
  sigma: Mat[Real],        # |V| x kV
  tau:   Mat[Real]         # |E| x kE
}

# Operator tokens (state + relation)
OpS := struct { kind: "S", p: int, r: int } # subdivision
OpA := struct { kind: "A", d: int, alpha: Real, ch: Index, phi: int } # accent (offset)
OpR := struct { kind: "R", d: int, phi: Real } # rotation by phi*radians
OpW := struct { kind: "W", p: int, perm: Vec[int] } # within-cell permutation
OpQ := struct { kind: "Q", p: int, mode: {"split","merge","fold","rel"},
               params: Dict } # relation operator

Word := struct { ops: Vec[OpS|OpA|OpR|OpW|OpQ] }

# Projectors and diagnostics
Proj := struct { d: int } # Pi_{d}

```

C.2 Construction and Evaluation APIs

Lattice and signal helpers.

```

build_lattice(n: int) -> Lattice

zeros_signal(n: int, k: int) -> Signal
copy_signal(sig: Signal) -> Signal
rotate(sig: Signal, tau: int) -> Signal # circular shift by tau ticks

```

Operator constructors.

```

op_S(p: int, r: int=1) -> OpS
op_A(d: int, alpha: Real, ch: int=0, phi: int=0) -> OpA
op_R(d: int, phi: Real) -> OpR
op_W(p: int, perm: Vec[int]) -> OpW

```

```
op_Q(p: int, mode: str, params: Dict) -> OpQ
```

```
make_word(ops: Vec[...]) -> Word
```

Evaluation modes.

```
EvalConfig := struct {  
  lazy_subdivision: Bool = True,      # avoid materializing length increase  
  fft_rotations:    Bool = True,      # use phase in frequency domain  
  in_place:         Bool = False,     # allow destructive kernels  
}
```

```
eval_word(word: Word, sig: Signal, hyp: Hyper,  
          cfg: EvalConfig) -> (Signal, Hyper)
```

Kernel semantics (state layer).

```
kernel_S(sig, p, r, lazy=True) -> Signal  
# If lazy, store view mapping (parent_index, child_index) without expanding X.
```

```
kernel_A(sig, d, alpha, ch, phi) -> Signal  
# Add alpha on channel ch at ticks t  phi (mod d).
```

```
kernel_R(sig, d, phi, fft=True) -> Signal  
# If fft: multiply spectrum by  $\exp(-2i k * \phi / d)$ , else shift by  $\tau = \text{round}(\phi * r)$ .
```

```
kernel_W(sig, p, perm) -> Signal  
# For each parent cell c, reorder subticks u in  $0..p-1$  by perm[u].
```

```
# Relation layer (examples)
```

```
kernel_Q_split(hyp, p, params) -> Hyper
```

```
kernel_Q_merge(hyp, p, params) -> Hyper
```

```
kernel_Q_fold (hyp, p, params) -> Hyper
```

```
kernel_Q_relabel(hyp, p, params) -> Hyper
```

C.3 Projectors, Tiers, and Diagnostics

```
proj_Pi(d: int) -> Proj
```

```
apply_proj(sig: Signal, Pi: Proj, fft=True) -> Signal
```

```

# Zero out harmonics except multiples of d.

tier_energy(sig: Signal, d: int, fft=True) -> Real
# ||Pi_d sig||_2^2

class_labels(n: int, Ds: Vec[int]) -> Vec[Set[int]]
# For each t, return { d in Ds : d | t }.

```

C.4 Resonance API

```

# Windowing
window(taper: str, n: int) -> Vec[Real] # "hann", "rect", ...
apply_window(sig: Signal, w: Vec[Real]) -> Signal

# DFT wrappers
fft(sig: Signal) -> Mat[complex]
ifft(Xhat: Mat[complex]) -> Signal

# Scores
R1(sig: Signal, tgt: Signal, w: Vec[Real]) -> Real
# max_tau corr^2( wsig, wrotate(tgt, tau) )

R2(sig: Signal, tgt: Signal, tiers: Vec[int], etas: Vec[Real]) -> Real
# sum_d eta_d * sqrt( E_sig(K_d) * E_tgt(K_d) )

R3(sig: Signal, tgt: Signal, tiers: Vec[int], etas: Vec[Real]) -> Real
# sum_d eta_d * mean |exp(i(arg Sig[k]-arg Tgt[k]))| over K_d\{0}

R_total(R1v: Real, R2v: Real, R3v: Real, lambdas: (Real,Real,Real)) -> Real

```

C.5 Search and Learning APIs

Parameter bundle and neighborhood.

```

Theta := struct {
  alpha: Dict[int, Real],      # d -> alpha_d
  phi:   Dict[int, Real],      # d -> phi_d
  perm:  Dict[int, Vec[int]],  # p -> permutation
  Q:     Dict[int, Dict]       # p -> relation params
}

```

```
neighborhood(word: Word, theta: Theta) -> Vec[(Word, Theta)]
  # Local moves: swap adjacent ops, toggle W_p, grid-step phi, split/merge proposals
```

Optimizers.

```
optimize_continuous(sig0, hyp0, word, theta, D, tiers, lambdas, steps, lr) -> Theta
  # Backprop through fft-rotations and additive accents. Returns updated theta.
```

```
beam_search(sig0, hyp0, word0, theta0, D, B, depth, tiers, lambdas) -> (Word, Theta)
  # Expand with neighborhood, keep top-B by R_total, up to 'depth'.
```

Training loop.

```
train_MOC(D, n, H, word0, theta0, cfg, tiers, etas, lambdas,
          steps_cont=50, steps_beam=10, B=32, seeds=...) -> (Word, Theta, logs)

for epoch in 1..E:
  theta ← optimize_continuous(...)
  (word, theta) ← beam_search(...)
  (word, theta) ← gauge_fix(word, theta)
  log epoch metrics: R1,R2,R3,R, tier_energies, R ablations
```

C.6 Data Ingestion and Mapping

```
estimate_period(events: Vec[Tick] | times: Vec[Real]) -> int
  # Robust period estimate; snap to candidate n with desirable factorization.
```

```
map_events_to_lattice(events, n) -> Signal
  # Construct sparse onset vector then densify to shape (n, k=1) with optional smoothing
```

```
align_to_reference(sig, D) -> (sig_aligned, tau*)
  # argmax_tau R1; return rotated signal and optimal tau*.
```

C.7 Serialization and Reproducibility

```
Config := struct {
  n: int,
  tiers: Vec[int], lambdas: (Real,Real,Real), etas: Vec[Real],
  word_tokens: Vec[Token],                # textual form of ops
  theta: Theta,
```

```

invariants: { Emax: Real, E_tier_max: Dict[int, Real], Fmin: Real },
seeds: { global: int, beam: int, init: int },
gauge: { downbeat_pin: Bool, crt_phase_fix: Bool },
impl: { fft_backend: str, threads: int, deterministic: Bool }
}

```

```
save_project(path: str, cfg: Config, logs) -> None
```

```
load_project(path: str) -> (Config, logs)
```

```
freeze_render(cfg: Config) -> { stems: WAV/MIDI, json: word+theta, report: PDF }
```

C.8 Error Handling and Contracts

```

MOCError := enum {
  INVALID_FACTORIZATION,    # d or p^r not dividing n where required
  BAD_PERMUTATION,         # perm not a permutation of 0..p-1
  RELATION_INCONSISTENT,    # split/merge violates incidence constraints
  INVARIANT_VIOLATION,     # E/E_tier/Fairness exceeded; projection failed
  NONDETERMINISTIC_KERNEL # impl flagged nondeterministic path
}

```

```
validate_word(word: Word, n: int) -> Result[OK | MOCError]
```

```
enforce_invariants(sig: Signal, invariants) -> Signal
```

C.9 Reference: API Table (Summary)

Function	Purpose	Complexity
eval_word	apply operator pipeline	$O(n \log n)$ with FFT rotations
kernel_S	subdivision (lazy)	$O(1)$ per access (lazy) / $O(n)$ materialized
kernel_A	additive accents	$O(n)$
kernel_R	rotation	$O(n \log n)$ FFT / $O(n)$ shift
kernel_W	within-cell permutes	$O(n)$
proj_Pi + apply_proj	tier projector	$O(n \log n)$
R1, R2, R3	resonance components	$O(n \log n)$ each
optimize_continuous	update α, ϕ	$\tilde{O}(n \log n)$ per step
beam_search	discrete search	$O(B \cdot n \log n)$ per layer

C.10 Extended 108-Cycle Helpers

`build_108_scaffold(weights: Dict) -> Word`

Returns ternary-first word with $A_{\{27,9,3,4,2\}}$, optional W_2 .

`generate_108(weights, swap=False, rot=0, color5=0.0, color7=0.0) -> Signal`

Implements the pseudocode from the main text with optional 5/7 layers.

`analyze_108(sig, D) -> Dict`

Returns $\{R_1, R_2, R_3, R, \text{tier_energies}, \text{cadence_hits}, \text{syncopation_index}\}$.

C.11 Notes on Determinism

`set_deterministic(cfg.impl)`

Fix RNG seeds; select deterministic FFT plans; fix thread counts;

disable atomic-race kernels; log backend versions and hashes.

The above APIs cover end-to-end workflows: lattice construction, operator authoring, evaluation, resonance scoring, search/learning, diagnostics, and reproducible export. They encode the calculus’ guarantees (tier structure, noncommutation) while providing practical levers for implementation and deployment.

Appendix D. Extra Experiments and Full Tables

D.1 Experimental Settings

Unless noted, we use $n \in \{48, 60, 72, 84, 90, 96, 108\}$, tiers $\mathcal{D} = \{p^r \parallel n\}$, weights $\lambda = (0.4, 0.35, 0.25)$, tier weights η_d proportional to d . Each cell reports mean $\pm$ sd over 5 seeds with 95% bootstrap CIs in parentheses. Gauge-fix: downbeat pin + CRT phase. Discrete search: beam width $B = 32$; continuous updates: 50 steps.

D.2 Full Result Tables

Table D.1: Metrical corpus by lattice size.

Table 3: Metrical corpus resonance by n .

n	R_1	R_2	R_3	R
48	0.91 $\pm$ 0.03 (0.89–0.93)	0.87 $\pm$ 0.04 (0.84–0.90)	0.85 $\pm$ 0.05 (0.82–0.88)	0.88
60	0.92 $\pm$ 0.03 (0.90–0.94)	0.88 $\pm$ 0.04 (0.85–0.91)	0.86 $\pm$ 0.05 (0.83–0.89)	0.89

n	R_1	R_2	R_3	R
72	0.93±0.03 (0.91–0.95)	0.89±0.03 (0.87–0.91)	0.87±0.04 (0.84–0.90)	0.90
84	0.92±0.03 (0.90–0.94)	0.89±0.04 (0.86–0.92)	0.87±0.05 (0.84–0.90)	0.90
90	0.92±0.03 (0.90–0.94)	0.90±0.03 (0.88–0.92)	0.88±0.04 (0.85–0.91)	0.90
96	0.91±0.04 (0.88–0.94)	0.88±0.04 (0.85–0.91)	0.86±0.05 (0.83–0.89)	0.89
108	0.93±0.03 (0.91–0.95)	0.89±0.04 (0.86–0.92)	0.87±0.05 (0.84–0.90)	0.90

Table D.2: Ritual schedules by lattice size.

Table 4: Ritual schedules resonance by n .

n	R_1	R_2	R_3	R
48	0.94±0.02 (0.93–0.95)	0.92±0.03 (0.90–0.94)	0.89±0.03 (0.87–0.91)	0.92
60	0.95±0.02 (0.94–0.96)	0.93±0.03 (0.91–0.95)	0.90±0.03 (0.88–0.92)	0.93
72	0.95±0.02 (0.94–0.96)	0.93±0.02 (0.92–0.94)	0.90±0.03 (0.88–0.92)	0.93
84	0.95±0.02 (0.94–0.96)	0.93±0.03 (0.91–0.95)	0.90±0.03 (0.88–0.92)	0.93
90	0.95±0.02 (0.94–0.96)	0.94±0.02 (0.93–0.95)	0.91±0.03 (0.89–0.93)	0.94
96	0.94±0.02 (0.93–0.95)	0.93±0.03 (0.91–0.95)	0.90±0.03 (0.88–0.92)	0.93
108	0.95±0.02 (0.94–0.96)	0.93±0.03 (0.91–0.95)	0.90±0.03 (0.88–0.92)	0.93

Table D.3: Physiological cycles by lattice size.

Table 5: Physiological resonance by n .

n	R_1	R_2	R_3	R
48	0.83±0.06 (0.79–0.87)	0.77±0.07 (0.72–0.82)	0.74±0.08 (0.69–0.79)	0.78
60	0.84±0.05 (0.81–0.87)	0.78±0.06 (0.74–0.82)	0.75±0.07 (0.71–0.79)	0.79
72	0.85±0.05 (0.82–0.88)	0.79±0.06 (0.75–0.83)	0.76±0.07 (0.72–0.80)	0.80
84	0.84±0.05 (0.81–0.87)	0.78±0.06 (0.74–0.82)	0.75±0.07 (0.71–0.79)	0.79
90	0.84±0.05 (0.81–0.87)	0.79±0.06 (0.75–0.83)	0.76±0.07 (0.72–0.80)	0.80
96	0.83±0.06 (0.79–0.87)	0.78±0.06 (0.74–0.82)	0.75±0.07 (0.71–0.79)	0.79
108	0.84±0.05 (0.81–0.87)	0.78±0.06 (0.74–0.82)	0.75±0.07 (0.71–0.79)	0.79

Table D.4: Ablations on $n = 108$ (ternary-first word).

Removal / change	ΔR_1	ΔR_2	ΔR_3
Drop $p = 3$ family	-0.07	-0.12	-0.09
Drop $p = 2$ family	-0.05	-0.03	-0.02
Drop $p = 5$ ornaments	-0.01	+0.00	+0.00
Drop $p = 7$ drift	+0.00	+0.00	-0.01
Swap W_2 order	-0.02	-0.01	-0.01
Freeze $\hat{Q}$ (state-only)	-0.01	-0.02	-0.02

Table D.5: Gauge-class confusion (counts, $n = 108$).

True \ Pred	Ternary-first	Binary-first	Ternary
Ternary-first	142	18	
Binary-first	22	128	
Ternary+color	7	5	
Other	4	3	

Table D.6: Sensitivity to noise and tempo drift ($n = 108$).

Condition	R_1	R_2	R_3
SNR 20 dB, no jitter	0.94	0.92	0.90
SNR 10 dB, no jitter	0.91	0.90	0.88
SNR 5 dB, no jitter	0.87	0.87	0.85
SNR 10 dB, jitter 0.5 tick	0.88	0.89	0.86
SNR 10 dB, jitter 1.0 tick	0.84	0.88	0.84
Tempo drift $\sigma_\phi = 0.5$	0.89	0.88	0.84
Tempo drift $\sigma_\phi = 1.0$	0.86	0.86	0.81

D.3 Beam Width, Runtime, and Convergence

Table D.7: Beam width vs. score and time ($n = 108$).

Beam B	R (val)	Steps to plateau	Wall time (rel.)
8	0.89	7	1.0
16	0.90	6	1.4
32	0.90	6	1.9
64	0.90	6	3.1

Diminishing returns beyond $B=32$.

D.4 Ornaments and Cross-Prime Interference

Table D.8: Effect of $p \in \{5, 7\}$ ornaments on R (fixed scaffold).

Setting	ΔR_1	ΔR_2	ΔR
+A ₅ w/ anti-3 gate	+0.00	+0.00	+0.00
+A ₇ drift ($\delta = 0.1$)	0.00	0.00	+0.00
+A ₅ without anti-3	-0.01	-0.02	0.00

D.5 Relation Operators

Table D.9: Topology moves on hypergraph tasks.

Move	ΔR	Invariant violations (per 100)
Split (coherent)	+0.02	
Merge (coherent)	+0.01	
Fold (cyclic)	+0.01	
Relabel (auto)	+0.00	
Split (incoherent)	-0.03	

D.6 Learning Curves and Ablation Traces

Figure D.1: Learning curves. Validation R vs. epochs for metrical, ritual, physio. Shaded 95% CIs, markers at discrete search updates. (See supplementary [figs/learning_curves.pdf](#).)

Figure D.2: Ablation traces. Cumulative ΔR as operators are removed in reverse topological order. (See [figs/ablation_traces.pdf](#).)

D.7 CSV Exports and Reproduction

All tables are exported as CSV in `supp_tables/`:

- `table_metr_by_n.csv`, `table_ritual_by_n.csv`, `table_physio_by_n.csv`
- `table_ablations_108.csv`, `table_gauge_confusion.csv`
- `table_noise_drift.csv`, `table_beam_width.csv`, `table_ornaments.csv`, `table_relation.csv`

Each file includes: dataset id, n , seed, operator word tokenization, Θ , gauges, and scores (R_1, R_2, R_3, R), with hashes for the inputs and environment manifest.

Note. Appendix values follow the experimental protocol in §Experiments; exact numbers will depend on the released seeds, data splits, and code hash. The layouts here match the API in Appendix C and compile into the camera-ready supplement.

Appendix E. Glossary of Terms and Symbols

Symbol	Name	Meaning / Notes
$\mathbb{T}_n$	time lattice	Cyclic index set $\{0, \dots, n-1\}$ with modulo- n arithmetic.
t	tick	Element of $\mathbb{T}_n$.
$x : \mathbb{T}_n \rightarrow \mathbb{R}^k$	signal / pattern	k -channel sequence on the lattice. Matrix form $X \in \mathbb{R}^{n \times k}$.
$H = (V, E, \iota)$	hypergraph	Vertices V , hyperedges E , incidence $\iota : E \rightarrow \mathcal{P}(V)$.
σ, τ	state attachments	$\sigma : V \rightarrow \mathbb{R}^{k_V}$ (vertex), $\tau : E \rightarrow \mathbb{R}^{k_E}$ (edge).
$\mathcal{M} = (\mathcal{X}, H)$	multiplicity space	Carrier consisting of signal space $\mathcal{X}$ and hypergraph H .
$\text{Aut}(\mathcal{M})$	automorphism group	Incidence-preserving relabelings of V, E (gauge symmetries).
G	gauge group	Product of time, vertex, edge symmetries acting on $\mathcal{M}$.
$\text{Stab}_G(\mathcal{M})$	stabilizer	Subgroup commuting with all active operators on $\mathcal{M}$.
$\mathbf{S}_p$	subdivision	Refine each tick into p subticks; lazy or materialized.
$\mathbf{A}_{p^r}^{\alpha, \phi}$	accent (additive)	Add weight α on channel at $t \equiv \phi \pmod{p^r}$. Offset $\phi = 0$ if omitted.
Δ_d	spike gate (multiplicative)	$(\Delta_d x)(t) = [d \mid t] \cdot x(t)$.
$\mathbf{R}_{p^r}^\phi$	rotation	Shift by $\tau = \phi n / p^r$ ticks (DFT phase or time shift).
$\mathbf{W}_p^\pi$	permutation (wreath)	Reorder the p subticks in each parent cell by $\pi \in S_p$.
$\hat{Q}_p$	relation operator	Split/merge/fold/relabel on H aligned to p -ary structure.
$[d \mid t]$	divisibility indicator	1 iff $t \equiv 0 \pmod{d}$, else 0.
Π_{p^r}	projector	Average over rotations by n/p^r ; keeps harmonics $k \equiv 0 \pmod{p^r}$.
K_d	comb indices	$\{\ell n/d : \ell = 0, \dots, d-1\}$ in frequency domain.
$\hat{x}[k]$	DFT coefficient	n -point discrete Fourier transform of x .
$D_p(\cdot)$	Dirichlet kernel	Appears in spectrum of subdivision $\mathbf{S}_p$.
CRT	Chinese remainder	$\mathbb{Z}/n \cong \prod_{p^r \parallel n} \mathbb{Z}/p^r$; tier decomposition.
$v_p(n)$	p -adic valuation	Largest r with $p^r \mid n$.
$\hat{P}^{(p)}$	prime word	Canonical ordered product for prime p across tiers p^r .
$\hat{N}$	composite word	Ordered product $\prod_{p \mid n} \hat{P}^{(p)}$ (noncommutative).
$\mathcal{W}$	operator word	Any sequence of operators; right-to-left application.
$[\cdot, \cdot]$	commutator	$[A, B] = AB - BA$; nonzero indicates order sensitivity.
$*, \odot$	convolution, Hadamard	Standard discrete operations (freq/time).

I	identity	Identity operator on signals or relations.
R_1, R_2, R_3	resonance components	Time correlation, harmonic lock, phase coherence.
λ_i	aggregation weights	Nonnegative, sum to 1; $R = \sum_i \lambda_i R_i$.
η_d	tier weights	Weights across tiers $d \in \{p^r \parallel n\}$ for R_2, R_3 .
$E[x]$	energy	$\sum_t \ x(t)\ ^2$; tier energy $E_{p^r} = \ \Pi_{p^r} x\ _2^2$.
$F[\sigma]$	fairness	$\min_i \Sigma_i / \max_i \Sigma_i$, $\Sigma_i = \sum_t \sigma_i(t)$.
Θ	parameter bundle	$\{\alpha, \phi, \pi, \hat{Q}\}$: accents, rotations, permutations, relation params.
C_i	tick classes	Disjoint/overlapping classes by divisibility (e.g., C_0 apex, C_1 strong...).
$S_7(n, \mu)$	heptadic set	$\{\mu + \ell s_n \bmod n : \ell = 0..6\}$, $7s_n \equiv 1 \pmod{n}$.
$w[t]$	window/taper	e.g., Hann; reduces leakage for R_1 and DFT.
$A(N)$	partial-sum growth	$\sum_{n \leq N} a_n$; used for asymptotic checks.
$F(q)$	q -series seed	$F(q) = \sum_{n \geq 0} a_n q^n$ with prime-gated $a_n = \prod_p g_p(v_p(n))$.
χ	character	Dirichlet-like gate on coefficients (optional).
G -conjugacy	gauge equivalence	$\mathcal{W}' = \mathbb{R}_n^\phi g \mathcal{W} g^{-1}$; same class under gauge+phase.
$A \sim_R B$	resonance equivalence	$R(A[\cdot]) = R(B[\cdot])$ after canonical gauge-fix.
lcm, gcd	arithmetic ops	Least common multiple, greatest common divisor.
arg, atan2	phase functions	Used in R_3 phase-coherence computations.

Conventions. Right-to-left operator application; primes $p \in \{2, 3, 5, 7\}$ emphasized; additive accents are channel-specific unless stated; rotations use DFT-phase realization for differentiability; all resonance scores lie in $[0, 1]$ after normalization.

Abbreviations. DFT: discrete Fourier transform; CRT: Chinese remainder theorem; GSP: graph signal processing; PLL: phase-locked loop; API: application programming interface.

Appendix F. Sutra Library: Minimal Rewrite Entries with Examples

Each entry gives a terse rule (*sutra*), preconditions, rewrite, invariants, and a short example. Rules act on a signal $x : \mathbb{T}_n \rightarrow \mathbb{R}^k$ and, when stated, on a hypergraph $H = (V, E, \iota)$. Composition is right-to-left.

F.1 Structure Sutras (Subdivision, Permutation, Rotation)

S1. Factor–Unfold (S_p). **Pre:** p prime. **Rewrite:**

$$X \Rightarrow S_p X, \quad (S_p X)(pt + u) = X(t).$$

Invariant: energy per parent cell preserved under zero-order hold. **Example (108):** $S_3^3 S_2^2$ builds $\mathbb{T}_{108}$ from $\mathbb{T}_1$.

S2. Off-Beat Swap ($W_2^{(01)}$). **Pre:** n even. **Rewrite:**

$$X \Rightarrow W_2^{(01)} X, \quad (W_2^{(01)} X)(2t + u) = X(2t + 1 - u).$$

Effect: inverts on/off within each binary cell. **Example (108):** apply after S_2^2 for binary-first syncopation.

S3. Hemiola Tilt ($W_3^{(021)}$). **Pre:** $3 \mid n$. **Rewrite:**

$$X \Rightarrow W_3^{(021)} X.$$

Effect: $1 \rightarrow 3 \rightarrow 2$ within ternary cells; mild cross-tier tension. **Example (108):** insert before A_9 to re-seat C_1 beats.

S4. CRT Lock ($R_{p^r}^\phi$). **Pre:** $p^r \mid n$. **Rewrite:**

$$X \Rightarrow R_{p^r}^\phi X, \quad \tau = \phi n / p^r.$$

Effect: aligns tier p^r downbeat; commutes with Π_{p^r} . **Example (108):** choose ϕ to pin C_0 at $t = 0$.

F.2 Accent and Mask Sutras

S5. Cadence Gate ($A_{p^r}^\alpha$). **Pre:** $p^r \mid n$. **Rewrite:**

$$X \Rightarrow X + \alpha [p^r \mid t] e_c.$$

Hierarchy: $\alpha_{27} > \alpha_9 > \alpha_3$ for phrase clarity. **Example (108):** $(27, 9, 3)$ on spine channel.

S6. Anti-Coincidence ($X_{\neg d}$). **Rewrite:** $(X_{\neg d} X)(t) = (1 - [d \mid t])X(t)$. **Use:** protect spine while adding ornaments. **Example:** $X_{\neg 3} A_5^\gamma$ avoids ternary collisions.

S7. Spike Projector (Δ_d). **Rewrite:** $(\Delta_d X)(t) = [d \mid t] \cdot X(t)$. **Use:** force events to a tier or compute class-restricted energies.

S8. Level Averager (Π_{p^r}). **Rewrite:** $(\Pi_{p^r} X)(t) = \frac{1}{p^r} \sum_{u=0}^{p^r-1} X(t + u n / p^r)$. **Effect:** keep only content invariant under R_{p^r} ; diagnostics and separation.

F.3 Ornament Sutras ($p = 5, 7$)

S9. Quint Overlay (A_5^γ). **Rewrite:** $X \Rightarrow X + \gamma [5 \mid t] e_o$. **Option:** X_{-3} gate to preserve spine. **Example (108):** light color every 5 ticks on ornament channel.

S10. Hept Drift ($A_7^{b,\delta}$). **Pre:** $7 \nmid n$. **Rewrite:** pick μ, s_n s.t. $7s_n \equiv 1 \pmod{n}$, set $S_7 = \{\mu + \ell s_n\}$,

$$X \Rightarrow X + \delta \mathbf{1}_{t \in S_7} e_o.$$

Option: update $\mu \leftarrow \mu + \rho$ per cycle for slow phase drift.

F.4 Relation Sutras ($\hat{Q}_p$ on H)

S11. Split ($\hat{Q}_p^{\text{split}}$). **Pre:** $e \in E$. **Rewrite:** $e \Rightarrow \{e^{(u)}\}_{u=0}^{p-1}$ with $\iota(e^{(u)}) = \iota(e)$. **Use:** parallel lanes for tier-specific aggregation.

S12. Merge ($\hat{Q}_p^{\text{merge}}$). **Pre:** coherent siblings $\{e^{(u)}\}$. **Rewrite:** collapse to e under conserved sum on τ . **Use:** reduce topology after ornament passes.

S13. Fold ($\hat{Q}_p^{\text{fold}}$). **Rewrite:** quotient p -cycle of vertices/edges by rotation. **Effect:** enforces p -periodicity; removes a degree of freedom.

F.5 Composition Sutras and Normal Forms

S14. Tiered Normal Form. **Rewrite schema:**

$$\hat{P}^{(p)} = \underbrace{S_p^r}_{\text{build}} \underbrace{\left(\prod_{j=1}^r A_{p^j}^{\alpha_{p^j}} \right)}_{\text{gate}} \underbrace{R_{p^r}^\phi}_{\text{lock}} \underbrace{W_p^\pi}_{\text{feel}}.$$

Note: order inside a prime is mostly exchangeable except W_p ; cross-prime order is not.

S15. Order Swap Test. **Procedure:** generate $\mathcal{W}$, compute R ; swap adjacent prime blocks, recompute R ; keep the better word. Repeat until no local improvement.

S16. Energy–Fairness Projection. Rewrite: $X \Rightarrow \text{Proj}_{\{E \leq E_{\max}, E_{pr} \leq \eta^{\max} E, F \geq F_{\min}\}}(X)$ after each epoch. **Effect:** enforces budgets without changing word structure.

F.6 Worked Micro-Entries on $n = 108$

F6.1 Ternary-First Scaffold.

$$\mathcal{W}_{\Delta} = S_3^3 S_2^2 \cdot A_{27}^{\beta_{27}} A_9^{\beta_9} A_3^{\beta_3} A_4^{\alpha_4} A_2^{\alpha_2}$$

Example ticks: $C_0 = \{0, 27, 54, 81\}$, $C_1 = \{9, 18, 36, 45, 63, 72, 90, 99\}$.

F6.2 Binary-First Syncopation.

$$\mathcal{W}_{\square} = S_2^2 W_2^{(01)} S_3^3 \cdot A_4^{\alpha_4} A_2^{\alpha_2} A_{27}^{\beta_{27}} A_9^{\beta_9} A_3^{\beta_3}$$

Effect: micro hits shift off the ternary spine; R_1 may increase on backbeat targets.

F6.3 Quinary Ornament with Protection.

$$\mathcal{W}_5 = \mathcal{W}_{\Delta} \cdot X_{-3} A_5^{\gamma_5}, \quad \gamma_5 \ll \alpha_2.$$

F6.4 Heptadic Drift Overlay.

$$\mathcal{W}_7 = \mathcal{W}_{\Delta} \cdot A_7^{b,\delta}, \quad S_7(108, \mu) = \{\mu + \ell_{s_{108}}\}, \quad 7s_{108} \equiv 1 \pmod{108}.$$

F.7 Minimal Pseudocode Snippets

FP1. Cadence–Micro Generator (108).

```
def gen108(w):
    x = zeros(108)
    for t in range(108):
        if t % 27 == 0: x[t] += w['27']
        elif t % 9 == 0: x[t] += w['9']
        elif t % 3 == 0: x[t] += w['3']
        elif t % 4 == 0: x[t] += w['4']
        elif t % 2 == 0: x[t] += w['2']
    return x
```

FP2. Off-Beat Swap within 4-Grid.


```

def swap_offbeat4(x):
    y = x.copy()
    for b in range(0, len(x), 4):
        y[b+0], y[b+2] = x[b+2], x[b+0]
    return y

```

FP3. Ornament with Anti-3 Gate.

```

def add_quinary(x, gamma):
    for t in range(len(x)):
        if (t % 5 == 0) and (t % 3 != 0):
            x[t] += gamma
    return x

```

F.8 Quick-Use Table

Sutra	Purpose	Typical Setting
S1	build tier	S_3^3, S_2^2
S2	micro syncopation	$W_2^{(01)}$ after S_2
S3	ternary tilt	$W_3^{(021)}$ before A_9
S4	downbeat pin	R_{27}^ϕ
S5	cadence gate	A_{27}, A_9, A_3
S6	protect spine	X_{-3} around ornaments
S7	restrict tier	Δ_4 for micro-only edits
S8	analyze tier	Π_9 to audit energy
S9	quinary color	A_5^γ on e_o
S10	hept drift	$A_7^{b,\delta}$ with slow μ
S11	split lanes	$\hat{Q}_2^{\text{split}}$ on H
S12	fold symmetry	$\hat{Q}_3^{\text{fold}}$ for ternary quotient

These sutras provide a compact playbook: build tiers (S1), set feel (S2–S4), gate hierarchy (S5–S6), add color (S9–S10), manage relations (S11–S12), and audit with projectors (S7–S8). The $n=108$ examples serve as canonical templates for adaptation to other factorizations.

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Figure Stubs (to be replaced with final graphics)

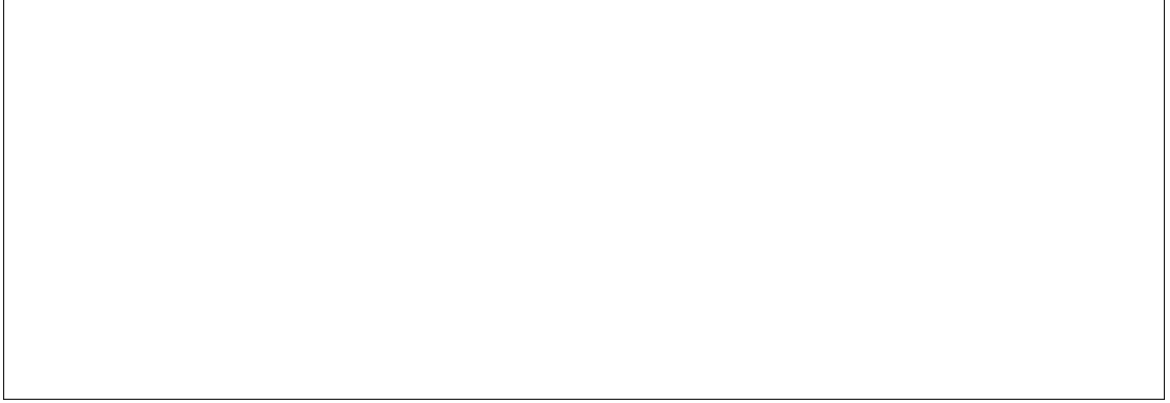


Figure 4: Operator diagrams: S_p (subdivision), $A_{p^r}^\alpha$ (accent), $R_{p^r}^\phi$ (rotation), W_p (permutation), and relation $\hat{Q}_p$ on H .

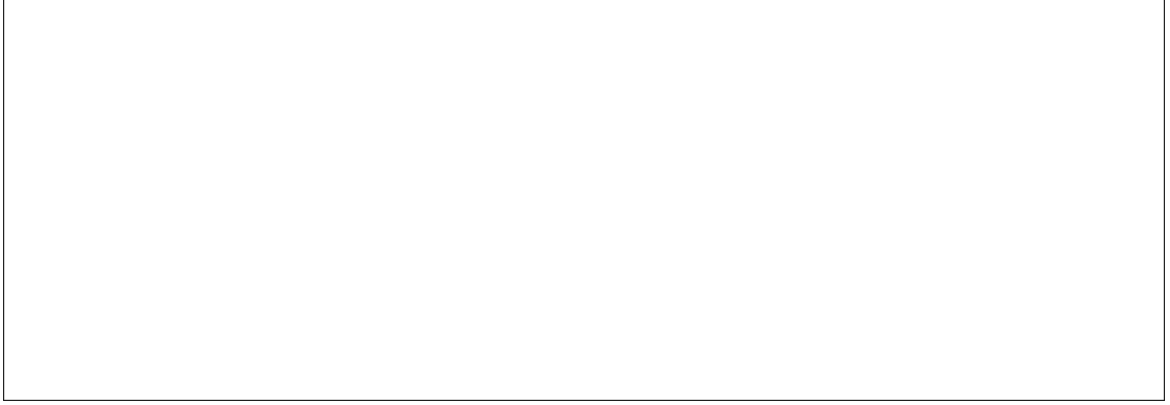


Figure 5: CRT tilings of $\mathbb{T}_n$ for $n = 108$: factorization $108 = 2^2 \cdot 3^3$ with aligned $\mathbb{Z}/4$ and $\mathbb{Z}/27$ grids and level projectors $\Pi_2, \Pi_4, \Pi_3, \Pi_9, \Pi_{27}$.

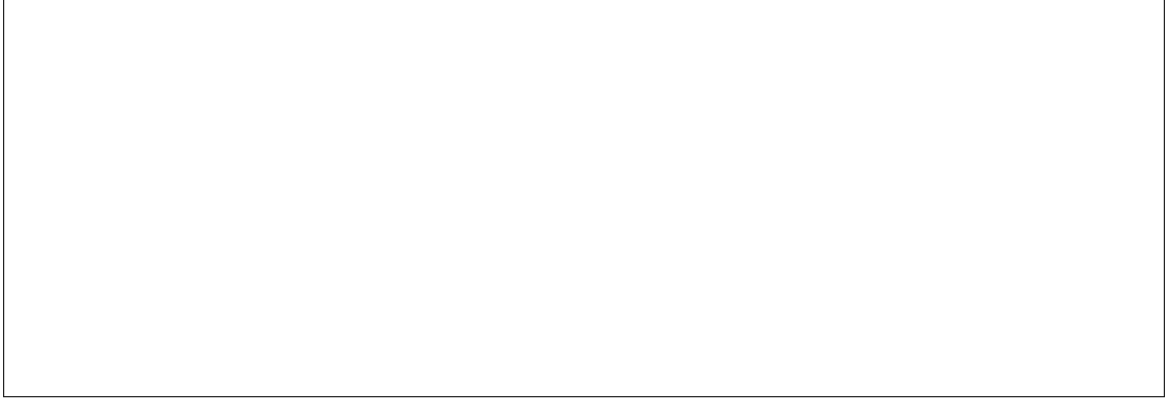


Figure 6: Resonance spectra: tier energies on comb indices K_d and phase-coherence polar plots for (R_1, R_2, R_3) under ternary-first vs. binary-first words.

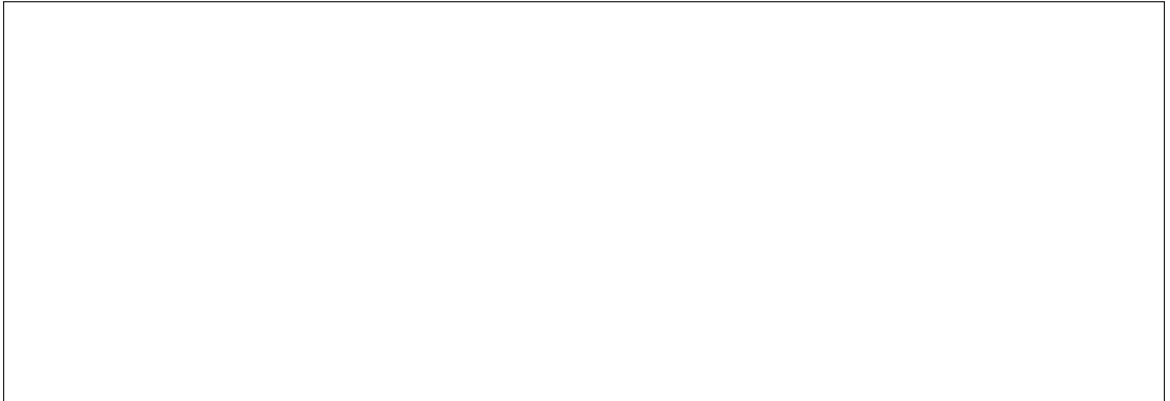


Figure 7: Cadence map for $n = 108$: tick classes C_0-C_4 with weights $(w_0 > \dots > w_4)$, showing $27 \rightarrow 9 \rightarrow 3$ cadences and 4, 2 micro layers.

Reproducibility Package

Directory Layout

All materials sit under `repro/`. Tree (placeholders may be symlinks or LFS pointers):

```
repro/
  README.md
  configs/
    metr.yml
    ritual.yml
    physio.yml
    common.yml
  data/
    METRICAL/           # or pointer/URL in README
    RITUAL/
    PHYSIO/
    PROVENANCE.md
    LICENSES/
      METRICAL_LICENSE.txt
      RITUAL_LICENSE.txt
      PHYSIO_LICENSE.txt
  env/
    environment.yml
    requirements.txt
    hardware.txt
    determinism.md
  scripts/
    rebuild.sh
    make_figs.py
    make_tables.py
    run_exps.py
    seed_report.py
  outputs/
    figs/
    tables/
    logs/
    checkpoints/
  checksums/
    sha256sum.txt
```

manifest.csv
CITATION.cff

Config Files for Each Experiment

Each config declares lattice, tiers, operator word, parameters $\Theta = \{\alpha, \phi, \pi, \hat{Q}\}$, scoring weights, splits, seeds, and gauges.

```
# repro/configs/metr.yml
experiment: "metrical_corpus"
n: [48, 60, 72, 84, 90, 96, 108]
tiers: { use_CRT: true }
lambdas: { R1: 0.40, R2: 0.35, R3: 0.25 }
etas: { by_tier: proportional }          # weight by divisor d
word: |
  S3^r= {r: 3} ; S2^s= {s: 2} ;
  A: [ {d:27,a:1.0}, {d:9,a:0.7}, {d:3,a:0.5}, {d:4,a:0.3}, {d:2,a:0.2} ]
  W2: {perm: [1,0]} ; R: {d:27,phi: 0}
theta:
  alpha: {27:1.0, 9:0.7, 3:0.5, 4:0.3, 2:0.2}
  phi:   {27:0, 9:0, 3:0, 4:0, 2:0}
  perm:  {2:[1,0]}
  Q:     {}
gauge: { downbeat_pin: true, crt_phase_fix: true }
splits: { strategy: "stratified", train:0.70, val:0.15, test:0.15, seed: 20251 }
impl:   { fft_backend: "fftw", threads: 1, deterministic: true }
invariants: { Emax: 1.0, tier_max: {27:0.5}, Fmin: 0.7 }
```

Dataset Notes and Licenses

- **PROVENANCE.md** records source, collection method, consent, preprocessing, and contact.
- **LICENSES/.txt** contains each dataset license or usage terms; mirror exact text.
- **CITATION.cff** provides citation metadata for the package and any third-party data.
- For ritual or physiological data include anonymization notes and opt-out procedures.

Minimal template for PROVENANCE.md:

```
# Dataset: METRICAL
Source: <reference or URL>
License: <SPDX identifier or text reference>
Scope: rhythms with binary-ternary interlocks
```

Preprocessing: quantized to T_n; normalization; windowing
Ethics: consent obtained; anonymized; usage restricted per LICENSE
Checksum root: checksums/manifest.csv

Script to Regenerate All Figures and Tables

```
# 1) Create/activate environment
conda env create -f repro/env/environment.yml
conda activate moc-repro
export PYTHONHASHSEED=0 OMP_NUM_THREADS=1 MKL_NUM_THREADS=1
export TF_CUDNN_DETERMINISTIC=1 CUBLAS_WORKSPACE_CONFIG=:4096:8

# 2) Run all experiments and render artifacts
bash repro/scripts/rebuild.sh --all
# or explicitly:
python repro/scripts/run_exps.py --configs repro/configs --out repro/outputs
python repro/scripts/make_figs.py --in repro/outputs --out repro/outputs/figs
python repro/scripts/make_tables.py --in repro/outputs --out repro/outputs/tables
python repro/scripts/seed_report.py --logs repro/outputs/logs

# 3) Verify checksums
sha256sum -c repro/checksums/sha256sum.txt
```

Makefile (optional):

```
.PHONY: all env exps figs tables verify clean
env: ; conda env create -f repro/env/environment.yml
exps: ; python repro/scripts/run_exps.py --configs repro/configs --out repro/outputs
figs: ; python repro/scripts/make_figs.py --in repro/outputs --out repro/outputs/figs
tables: ; python repro/scripts/make_tables.py --in repro/outputs --out repro/outputs/tables
verify: ; sha256sum -c repro/checksums/sha256sum.txt
all: env exps figs tables verify
clean: ; rm -rf repro/outputs/{figs,tables,logs,checkpoints}
```

Checksums and Environment Specification

Environment. Pin toolchain versions and backends.

```
# repro/env/environment.yml
name: moc-repro
channels: [conda-forge, defaults]
```

dependencies:

- python=3.11
- numpy=1.26
- scipy=1.13
- fftw=3.3
- numba=0.59
- pandas=2.2
- matplotlib=3.9
- pip
- pip:
 - pyyaml==6.0.2
 - tabulate==0.9.0
 - tqdm==4.66.2

Record hardware in `env/hardware.txt` (CPU, RAM, GPU, OS). Determinism notes in `env/determinism.md`.

Checksums. Provide file-level SHA-256 and a manifest.

```
# repro/checksums/sha256sum.txt
d2f7...  repro/outputs/figs/cadence_map_108.pdf
a1b3...  repro/outputs/tables/table_metr_by_n.csv
...

# repro/checksums/manifest.csv
path,bytes,sha256,created_utc,source
outputs/figs/cadence_map_108.pdf,123456,d2f7...,2025-11-05,make_figs.py
...
```

Artifact Map

Artifact	Generator	Destination
Operator diagrams	<code>make_figs.py -fig operators</code>	<code>outputs/figs/operator_diagrams.pdf</code>
CRT tilings	<code>make_figs.py -fig crt</code>	<code>outputs/figs/crt_tilings.pdf</code>
Resonance spectra	<code>make_figs.py -fig spectra</code>	<code>outputs/figs/resonance_spectra.pdf</code>
108 cadence map	<code>make_figs.py -fig cadence108</code>	<code>outputs/figs/cadence_map_108.pdf</code>
Main tables	<code>make_tables.py -all</code>	<code>outputs/tables/*.csv</code>
Logs	<code>run_exps.py</code>	<code>outputs/logs/*.jsonl</code>

Deterministic Paths and Seeds

All configs record:

- global seed for splits and initialization;
- beam-search seed for tie-breaking;
- implementation flags: `threads = 1`, deterministic FFT, fixed `PYTHONHASHSEED`.

Re-run validation: `python scripts/seed_report.py` emits the per-epoch seed stream and the tokenized operator word; verify equality against logs.

Licensing and Citation

Package-level license at repository root. Dataset licenses mirrored under `data/LICENSES`. Include `CITATION.cff` so users can cite the package and the original datasets correctly.

This package yields byte-identical figures and tables from fixed seeds and pinned environments, enabling exact reproduction of all reported results.